

# Development and Testing of Machine Learning Framework for Effective Application on Microbiome-Based Classification Problems

A thesis submitted to the University College Cork  
in partial fulfilment of the requirements for the degree of  
Master in Bioinformatics and Computational Biology  
in collaboration with SeqBiome.

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September 2024

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## LIST OF ABBREVIATION

| Abbreviation | Definition                                      |
|--------------|-------------------------------------------------|
| GM           | Gut Microbiome                                  |
| PCR          | Polymerase Chain Reaction                       |
| NGS          | Next-Generation Sequencing                      |
| PCoA         | Principle Coordinate Analysis                   |
| OUT          | Operational Taxonomic Unit                      |
| ASV          | Amplicon Sequence Variant                       |
| RSV          | Ribosome Sequence Variant                       |
| TSS          | Total Sum Scaling                               |
| CSS          | Cumulative Sum Scaling                          |
| TMM          | Total Mean of M-Values                          |
| ALR          | Additive Log-Ratio                              |
| CLR          | Centre Log-Ratio                                |
| ILR          | Isometric Log-Ratio                             |
| PCA          | Principle Component Analysis                    |
| mRMR         | Maximum-Relevance Minimum-Redundancy            |
| RFE          | Recursive Feature Elimination                   |
| LASSO        | Least Absolute Shrinkage and Selection Operator |
| SVM          | Support Vector Machines                         |

|      |                                                 |
|------|-------------------------------------------------|
| GLM  | Generalised Linear Model                        |
| K-NN | K-Nearest Neighbours                            |
| ANN  | Artificial Neural Network                       |
| CV   | Cross-Validation                                |
| SHAP | SHapley Additive exPlanations                   |
| LIME | Local Interpretable Model-agnostic Explanations |
| PIMP | Permutation Importance                          |
| EC   | Enzyme Commission                               |
| AUC  | Area Under Curve                                |
| ROC  | Receiver Operating Characteristic               |
| IQR  | Interquartile Range                             |
| SCFA | Short-Chain Fatty Acid                          |

## **ACKNOWLEDGEMENTS**

I would like to express my gratitude to my family and friends for their support throughout this year. Your encouragement has been invaluable to me. I want to extend particular thanks to Niamh Hurley, who provided critical technical support during the project.

I am grateful to my supervisors for their feedback and guidance during both the research and writing phases of this thesis. Your patience has been helpful in shaping this work.

Finally, a thanks to my classmates for the support and I congratulate each of you on your success.

## ABSTRACT

The human gut microbiome plays an important role in various physiological processes, including athletic performance. Traditional methods such as principal coordinate analysis and differential abundance analysis face limitations in handling the complexity and high dimensionality of microbiome data. This study explores the application of machine learning techniques to address these challenges, using the athletic gut microbiome as a model system. Our findings highlighted the importance of comprehensive parameter tuning and model selection in ML applications to capture biologically meaningful patterns. A review of statistically significant features from the predictive models of both 16S rDNA sequencing data of marathon runners and shotgun metagenomic sequencing data of various athletes, were investigate. The results indicate that the athletic gut microbiome is dominated by species from the *Faecalibacterium* genus and other butyrate producers, suggesting a dynamic microbial ecosystem rather than individual species driving athletic performance. Furthermore, novel species were identified that had not been previously reported in this context. These findings offer new insights into the role of short-chain fatty acid production and microbial metabolic pathways in athletic performance. Future research should focus on identifying unknown species and further exploring their metabolic contributions, particularly in relation to short-chain fatty acid synthesis.

## **AIMS**

The study is driven by the growing recognition of the human gut microbiome's role in influencing various physiological processes, including athletic performance. Despite advances in microbiome research, studies that investigate effective application of machine learning techniques to classify and interpret microbiome data remain few. The aim of this study is to develop and test a comprehensive machine learning framework that can accurately analyse microbiome data and identify bacterial species associated with athletic performance. The study makes use of data from long-distance marathon runners and publicly available shotgun metagenomics datasets derived from the faecal samples of healthy athletes to achieve this goal. By focusing on both the methodological aspects of machine learning and the specific characteristics of microbiome data, this research seeks to bridge gaps in current understanding and provide valuable insights into how gut microbiota can impact athletic outcomes.

# INTRODUCTION

## Athlete Microbiome

The gut microbiome (GM) is the collection of genetic information that provides information about the present microbiota (collection of microorganisms) and the functional capacity of the ecosystem within the gut (Hughes and Holscher, 2021). It consists of approximately 100 trillion microbial cells, representing roughly 9 million genes. Most of the microbial cells are bacteria but archaea, fungi (generally *Candida* and *Saccharomyces* species), protozoans and viruses are also present. The bacterial cells can generally be grouped into 5 phyla and 160 species with Bacteroidetes and Firmicutes being the most dominant populations followed by Actinobacteria, Proteobacteria and Verrucomicrobia (Clark and Mach, 2016; Wosinska *et al.*, 2019).

The human GM has undergone extensive scrutiny and study to determine the role of the GM on health, digestion, and disease prevention and treatment. As such, the GM has been found responsible for the synthesis of vitamins, oxidative stress control, motility of the intestines, inflammatory modulation and host immune response (by neutralising drugs and carcinogens, protecting against pathogens and stimulating and maturing immune and epithelial cells), control of excitatory and inhibitory neurotransmitters (in response to physical or emotional stress) and promotion of digestion and host energy production (through food and nutrient absorption) (Clark and Mach, 2016; Mohr *et al.*, 2020). Digestion and energy production are modulated by the digestion and fermentation of carbohydrates to SCFAs (Short Chain Fatty Acids) such as acetate, 36%; propionate, 9%; butyrate 2% by microbiota (Hughes and Holscher, 2021).

Recently, there has developed interest towards the GM and exercise, physical activity and athlete performance. However, it is important to distinguish between exercise and physical activity when investigating the interactions between the GM and athletes/sports. Exercise is a subgroup of physical activity that involves a structured repetitive movement with a final goal of achieving improvement or maintenance of physical fitness (Hughes and Holscher, 2021). This is more in line with the actions of athletes than simply expending energy through physical movement, known as physical activity. Exercise itself can be broadly categorised into either static (e.g. weightlifting) or dynamic (e.g. marathon running). Dynamic exercise is often associated with an enrichment of positive bacterial species as opposed to static exercise (O'Brien *et al.*, 2022). The reverse has also been found to be true, as Chen *et al.*, discovered that introducing probiotics to the gut

microbiota of mice enhanced their endurance, suggesting potential performance benefits for athletes.

Most studies investigating the impact of exercise on the GM and its composition have been conducted through animal studies. Through these animal studies, exercise has been found to be involved in the modulation of high-fat-induced obesity, cardiovascular disease, and colonic epithelial cell proliferation. It also promotes gut barrier integrity and regulates the host immune system and gene expression (Jang *et al.*, 2019; Zhou *et al.*, 2022). These effects are attributed to the increase in butyrate metabolites and butyrate-producing bacteria, which provide essential fuel for colonocytes (Mailing *et al.*, 2019).

Animal studies are often preferred because they allow for stricter control of confounding variables like nutrition, which is closely linked to exercise, whereas controlling these variables in human studies is more challenging (Hughes and Holscher, 2021). In contrast, research investigating the effects of exercise in humans is limited, and studies focused on athletic performance have only started to emerge in recent years (Giron *et al.*, 2022). Research on marathon runners has shown differential enrichment of gut microbiota with respect to specific bacteria, such as *Veillonella atypica*, compared to non-runners. This study also found that lactate to propionate metabolising genera were found in higher abundance postexercise when examining a cohort of elite athletes, suggesting that promoting the conversion of exercise-induced lactate into propionate could enhance athletic performance by improving heart rate, VO<sub>2</sub> max, and resting energy expenditure (Scheiman *et al.*, 2019). Additional research has highlighted strong correlations between butyrate-producing bacteria and VO<sub>2</sub> max, associations of microbial short-chain fatty acid (SCFA) pathways with athletic performance in rugby players, and evidence that the athlete lifestyle actively shapes the GM, favouring bacterial species with higher enzymatic potential, which can positively impact health and muscular performance (Estaki *et al.*, 2016; Barton *et al.*, 2018; Fontana *et al.*, 2023).

In general exercise has been shown to be a strong determinant of health status. The GM has been shown to alter energy metabolism, immune function and oxidative stress, all of which are vital for athlete's performance and overall health. However, not enough human studies have been conducted to draw adequate conclusions, therefore, necessitating the need for further research into the interaction between the GM and exercise. As the understanding of this complex

relationship deepens, the potential for optimising athletic performance through targeted interventions in the gut microbiome may emerge in the future.

## **NGS Techniques and Analysis**

As mentioned above, research on the GM and its microbiota is a relatively new field that has gained momentum with the development of advanced DNA sequencing technologies and computational methods. These advancements enable the dramatic increase of scale, scope and ability to manage the mass information density of the GM (Sidebottom, 2023). Previously, the study of the human GM was significantly more difficult as it heavily relied on the ability to culture microbes in a laboratory environment. However, this method has significant limitations as gut microbes have an extremely specific set of growth conditions *in vivo* that are difficult to recreate in a lab environment (Rankin *et al.*, 2017).

Amplicon sequencing, often known as 16S rRNA gene sequencing because it is the most common target for PCR sequencing, is one of the most widely used Next-Generation Sequencing (NGS) methods for the GM identification and characterisation as it is significantly more cost-effective and scalable to larger projects than other approaches (Wensel *et al.*, 2022). Other frequently used forms of data from NGS include; metagenomic data, used to generate detailed characterisation of microbial gene structures and their functionality; and metatranscriptomics, used to profile differential microbial gene expression (Mohr *et al.*, 2020). Metaproteomics and metabolomics are alternative data forms of NGS but are less frequently used due to the presence of many unknown proteins and metabolites in their respective databases. Additionally, the lack of a unique protocol further limits their widespread application (Rezasoltani *et al.*, 2020). These data types each provide their own unique investigation into the interaction mechanisms between microbiota and the microbiome itself. Therefore, integrative analysis is crucial to deepen our understanding of the GM.

While 16S rRNA gene sequencing is the most commonly used NGS, it is slowly being replaced by whole-community shotgun metagenomic analysis as the price gap between the two is lowered. This is because shotgun metagenomics provides a higher resolution and sensitivity towards characterisation at the genus and species level (Mohr *et al.*, 2020).

A variety of statistical analyses can be performed to assess the large data produced from high-throughput NGS. Many studies have measured alpha diversity, a measure of biological diversity

in a sample or environment (e.g. richness, evenness); beta diversity, a measure of how different communities are between samples; PCoA which identifies clustered patterns between samples; and differential abundance analysis, which aims to find the differences in the abundance of taxa between two groups, conditions or timepoints by assigning a significance score to the comparison (Kumar *et al.*, 2014; Cappellato, Baruzzo and Di Camillo, 2022). However, each of these cannot properly capture the complex interactions within the data as they may be highly impacted by the dimensionality and scale of the data, resulting in important information loss and bias introduction (Ghannam and Techtmann, 2021). Therefore, this highlights the need for machine learning in microbiome analysis, as it can adequately account for the variation in the microbiome and typically assess the effects of each bacterial population separately (Topçuoğlu *et al.*, 2020). However, developing a machine learning framework can prove challenging due to the inherent complexity of microbiome data. The process requires careful consideration of each step of the framework, from data handling to visualisation of results. This will be elaborated upon in the following sections.

## **Preprocessing Techniques for Machine Learning**

Data preprocessing is the first step involved in any machine learning pipeline. Real-world data often contains issues that can significantly influence the generalised performance of predictive models, making preprocessing an essential step. These issues are often divided into three categories; excess data, which adds complexity leading to overfitting and reduced interpretability; limited/missing data, which reduces model accuracy and introduces bias; and fractured data, which consists of incompatible data for downstream analysis (Gygi, Kleinstein and Guan, 2023; Kang, 2013; Maharana, Mondal and Nemade, 2022).

Data preprocessing attempts to remedy these issues by transforming raw data, through a series of steps, into a state in which supervised machine learning algorithms can quickly interpret the data and perform statistical analysis. These steps generally involve data cleaning, integration, transformation, and reduction (Malley, Ramazzotti and Wu, 2016).

### **Data Cleaning**

Data cleaning is an essential step of preprocessing that attempts to handle the issue of missing values but does little to deal with the issue of limited data. Discarding and imputing missing values are the two most frequently used data cleaning methods (Kwak and Kim, 2017). There are two methods of discarding missing data, complete case analysis and available case analysis. Complete case analysis is the most frequently used method and involves discarding all instances

with missing values, with analysis being carried out on the remaining data. It is a simplistic method but reduces the sample size and the statistical power of the model. Available case analysis uses all available data between corresponding variables which allows for a greater sample size (Jamshidian and Mata, 2007). However, this method isn't as simple and can cause various sample sizes between variables.

Imputation of missing data involves substituting the values with an estimation based on information from the inferred population parameters using statistical analysis (Roy *et al.*, 2018). It often does not provide the most accurate prediction for the missing values but can certainly help improve the completeness of the data. There are various imputation approaches (e.g. multiple imputation), each with their own distinctive methods (e.g. regression method), however, there is no consensus in the literature regarding an ideal approach as they all contain a set of advantages and limitations (Zhang, 2012; Jäger, Allhorn and Bießmann, 2021).

### **Data Integration**

Data integration is often used within systems biology to integrate different biological data such as gene expression and pathway data. From the perspective of systems biology, there are three levels in which data integration can be achieved; the data level, where data is adjusted and combined; the processing level, where correlations between datasets are identified; and the decision level, where results are mapped using prior knowledge (Lenzerini, 2002; Roy *et al.*, 2018).

### **Data Transformation**

Data transformation converts and structures the values within the dataset into a more suitable format that can be used by the predict model to enhance its interpretability (Malley, Ramazzotti and Wu, 2016). There are numerous transformation methods, each suited to specific types of data (Manikandan, 2010). Additionally, data transformation often involves the alternation of features into a common scale or unit, to balance the model's sensitivity between high magnitude features and low magnitude features. This is commonly known as data normalisation (Maharana, Mondal and Nemade, 2022). There are numerous normalisation algorithms; however, two are more widely used in the literature compared to the others: Min-max normalisation and Z-score normalisation. (Cao, Stojkovic and Obradovic, 2016). Min-max normalisation has been used in several biological studies to scale gene values within an interval between 0 and 1 (Osama,

Shaban and Ali, 2023). Z-score normalisation rescales the data so that the resultant features have a mean of zero and a standard deviation of 1 (Singh and Singh, 2020).

### **Data reduction**

Data reduction aims to reduce the original dataset into a more concise representation while preserving its integrity. This practice is crucial for handling large datasets, as complex analyses can be computationally intensive and impractical without reduction (Malley, Ramazzotti and Wu, 2016). A common approach to data reduction is to remove predictors with a single or few unique values. These are known as (near-)zero variance predictors, which are uninformative predictors that provide little to no benefit to the predictive model (Kuhn and Johnson, 2013). Some models are capable of handling these predictors (e.g. decision trees), however, other models (e.g. GLMs) can find this type of data problematic. Therefore, it is common practice to remove (near-)zero variance predictors as they are likely to decrease the computational complexity of the data, without compromising the performance. Additionally, data reduction can be achieved through feature selection and extraction, which will be discussed further in a later section.

By following these series of steps, it is possible to address the three main issues encountered when collecting real-world data: excess, limited, and fractured data. This process results in a dataset that can provide compatible and interpretable information for a predictive model to perform statistical analysis effectively. However, in microbiome data analysis, there are often additional steps involved in preprocessing used to treat the greater complexity inherent in microbiome data.

### **Preprocessing Techniques of Microbiome data**

In recent decades, Next-Generation Sequencing (NGS) technologies has led to a large-scale increase in microbiome datasets, often through either 16S rRNA sequencing or metagenomic sequencing of whole microbial genomes. Following initial quality control steps (e.g. trimming, chimera detection, etc), the sequencing data is typically presented into a large matrix with columns representing samples and rows representing either Operational Taxonomic Units (OTUs), Amplicon Sequence Variants (ASVs), or Ribosomal Sequence Variants (RSVs) (Callahan, McMurdie and Holmes, 2017; Jasner *et al.*, 2021). Several features involved with these large matrices can cause problematic results in downstream analysis if not carefully considered. The most prevalent issues of these features include, but are not limited to, differential sequencing

depth, zero-inflation, compositionality, and high dimensionality (Banks and Fienberg, 2003; Cho *et al.*, 2023; Morton *et al.*, 2017).

Numerous efforts have been made to establish a standardised method of preprocessing for microbiome data that attempts to mitigate the challenges mentioned above (Weiss *et al.*, 2017; Jasner *et al.*, 2021; Papoutsoglou *et al.*, 2023). Despite this, consensus on the optimal approach remains elusive, as each method has its own unique set of characteristics that justify its usage. There have been numerous normalisation techniques that have been developed for handling large count matrices from NGS, with rarefaction being one of the most commonly used normalisation techniques for microbiome data (Lozupone *et al.*, 2011; Wong, Wu and Gloor, 2016; Zmora *et al.*, 2018).

### **Rarefaction**

Rarefaction standardises the library size across all samples by randomly removing reads from each sample until a predetermined number is reached, ensuring uniform read counts. Despite its widespread use, there is ongoing debate over its reliability and accuracy. Some argue against its practical use, stating it should never be used in practice as it discards data, reducing the statistical power (McMurdie and Holmes, 2014). Conversely, others suggest that rarefaction can be viable depending on the extent of the data loss, with diminishing statistical power as more data is discarded (Weiss *et al.*, 2017). This leaves researchers with a difficult decision regarding the trade-offs between sampling depth and the number of samples evaluated.

### **Scaling Transformations**

The majority of other normalisation methods are scaling methods, which use a sample specific scale factor (e.g. total sum of counts in a single sample) to alter the data to account for differences in total counts or sequencing depths between samples (Pereira *et al.*, 2018). The specific effects of the scaling factor are dependent on the method, with quantile normalisation, median normalisation, Total Sum Scaling (TSS), Cumulative Sum Scaling (CSS) and Total mean of M-values (TMM) being the most frequently used methods (McKnight *et al.*, Lin and Peddada, 2020, 2019, Wang, Sun and Luan, 2024).

In particular, TSS is the most widely used and cited normalisation technique in microbiome research as it was adopted in its early development along with rarefaction. However, recent literature heavily criticises both scaling normalisation and rarefaction for their failure to address

the compositional nature of microbiome data, which is a prominent and defining characteristic (Xia, 2023). This is because when microbiome data is converted to relative abundances, normalised counts or rarefied, there is a loss of absolute abundance information which is crucial for understanding the biological significance of the data (Gloor *et al.*, 2017). Analysis of such data with the assumption of absolute abundance can lead to erroneous results for several common techniques in downstream analysis.

First, statistical models that assume independence between features are flawed because compositional data violates the assumptions of independence as one change in abundance inherently drives changes in another OTU. Therefore, it is impossible to determine if the growth or decline of a specific OTU has truly occurred (Morton *et al.*, 2017). Second, statistical analysis that uses distances between samples are misleading and erratically sensitive to the arbitrary inclusion or exclusion of components (Quinn *et al.*, 2019). Third, correlation analysis of compositional data can lead to meaningless conclusions as statistically independent components can appear highly correlated (Lovell *et al.*, 2015).

### **Aitchison Transformations**

To address these compositionality issues, it is recommended to use compositional transformations. These transformations utilise the concept of balances, which infer changes in the balance between specific subsets of the community, rather than changes in absolute abundance (Morton *et al.*, 2017). The concept of balance was originally proposed by Aitchison, that the underlying relationship between OTUs can be explained by treating the data as ratios (Aitchison, 1982). Additionally, Aitchison proposed the additive log-ratio (ALR) to address the issue of compositionality, by taking the data from the simplex to a real space, avoiding the pitfalls associated with constrained data (Morton *et al.*, 2017). However, ALR uses a reference OTU which must be pre-specified prior to transformation. To avoid predefining a reference OTU, Aitchison later proposed an alternative method which uses the centre of mass of all OTUs as reference instead. Therefore, with each sample, for each OTU, log-ratios are computed in relation to the geometric mean (Lin and Peddada, 2020). This is known as the centre log-ratio (CLR) transformation.

The isometric log-ratio (ILR) transformation is an alternative method later proposed by followers of Aitchison, which claims to overcome the shortcomings of ALR and CLR by associating angles and distances from the simplex to the real space (Egozcue *et al.*, 2003). Recent studies have

suggested the use of ALR as the preferred method as it is easier to interpret than CLR or ILR. However, in the absence of a reference or baseline OTU, CLR is a more balanced representation and should be used instead (Ibrahimi *et al.*, 2023).

## **Imputation**

A challenge presented when using log-ratio transformations is the prevalence of zero values in the data (zero-inflated data). Taking the logarithm of zero values results in an infinitely negative value, distorting the data and invalidating statistical analysis. Therefore, it is necessary to account for the sparsity in the data by imputing zero values. Multiple approaches have been proposed for imputation that claim to tackle the problem posed by zero values in microbiome data (Silverman *et al.*, 2020). However, one must be cautious when implementing imputation, as there is no clear consensus to the most effective approach.

Pseudo count addition is the most common and straightforward approach but can add bias that inflates low abundance OTUs in the data (Silverman *et al.*, 2020; Ibrahimi *et al.*, 2023). The selection of a pseudo count value is not based on rigorous statistical theory but is instead arbitrary (Mandal *et al.*, 2015). A common approach to pseudo count selection is to add the constant 1 to all values but alternative methods have been proposed which attempt to reduce the bias. The “const” and “unif” method implemented by Lubbe, Filzmoser and Templ, claims to minimise the bias in the covariance structure and reduce the distortion in the multivariate distribution, respectively (Lubbe, Filzmoser and Templ, 2021).

More sophisticated approaches have been developed to address the issues of zero-inflation by modelling the count distribution of the data. Zeros values in microbiome data can be divided into three categories; biological zeros or real zeros which represent the true absence of an OTU within the sample, and technical zeros which appear as zero values due to limitations in sequencing techniques and depth (Zhou *et al.*, 2023). Both over-dispersed and mixture count models have been proposed to treat zero-inflated data. These differ in the way they consider zero values. Over-dispersed models inherently assume that all zeros are technical zeros and do not consider the possibility of biological zeros (Yang and Chen, 2022). This assumption holds true for low abundance OTUs commonly found but proves inadequate when zero values are considered for OTUs that are often found abundant.

In contrast, mixture models include an extra parameter that accounts for the structural nature of zeros in the data. They are capable of distinguishing between both biological and technical zeros, imputing only technical zeros (Yang and Chen, 2022). This creates a more flexible model but at the cost of increased computational demand. There is an ongoing debate regarding the appropriateness of mixture-models. Some claim that by mixing two or more distributions and distinguishing between zero types, a more nuanced representation of microbial abundance is achieved (Zeng *et al.*, 2023). While others are arguing that mixture models reduce the statistical stability and power in downstream analysis, particularly when there are too many or too few zero values in the count data (Lyu *et al.*, 2023). Silverman *et al.*, recommends the approximation of biological zeros to sampling zeros, which is considered logical since in most cases this assumption will be made for OTUs which have small/negligible abundances. Therefore, overdispersion models were suggested for use (Silverman *et al.*, 2020).

However, Zhou *et al.* (2023) argues in a recent paper that parametric models, such as over-dispersed models are unlikely to fully capture the complex patterns in the data due to their inherent distributional assumptions. Thus, it recommends the usage of non-parametric imputation methods that make fewer assumptions and have been more widely used outside of microbiome and compositional data, like k-NN and Random Forest imputation.

Yet, for a large proportion of zeros which is commonly found in microbiome data, non-parametric methods like k-NN are likely to fail as there are too few neighbours for the algorithm to select from (Lubbe, Filzmoser and Templ, 2021). Thus, it may be recommended to use a parametric test that is specifically designed for compositional data. Bayesian Multiplicative Replacement maintains the compositionality and ensures that the sparsity of the data is handled in a biologically meaningful way (Palarea-Albaladejo and Martín-Fernández, 2015). In my opinion the consensus of imputation methods for zero counts is still an open and ongoing problem that requires further investigation before an adequate conclusion can be reached.

### **Summary of Data Transformation**

While the challenge of zero-inflation data has not been adequately addressed in literature, there still have been numerous papers in recent years that have tried to emphasise the importance of accounting for compositional data (Lovell *et al.*, 2015; Erb and Notredame, 2016; Gloor *et al.*, 2017; Morton *et al.*, 2017; Quinn *et al.*, 2019). And while the awareness is growing and an increasing number of publications using log-ratio transformations, these methods cannot yet be

considered a mainstream concept in microbiome analysis (Randolph *et al.*, 2018; Jasner *et al.*, 2021; Ibrahimi *et al.*, 2023).

Most of the characteristic features of large matrices with count data from microbiome communities will be accounted for by using Aitchison's log-ratio transformations along with an adequate imputation method. Differential sequencing depth is inherently handled by Aitchison's log-ratio transformations by adjusting for the differences in sequencing depth/library size when converting the counts into log-ratios. Additionally, the issue of compositionality is avoided by moving the data from the simplex to a real space. Aitchison's methods inherently require non-zero values, necessitating the use of a suitable imputation technique. Consequently, zero-inflated data is handled under the assumption that an appropriate imputation method has been properly applied. Therefore, the only prevalent characteristic feature of large matrices of count data that remains is high dimensionality.

### **High Dimensionality**

High dimensionality in large matrices of count data introduces several challenges due to the curse of dimensionality. One of the main issues is interpretability, as humans struggle to visualise high-dimensional spaces and calculate meaningful statistics (Armstrong *et al.*, 2022). This complexity is exacerbated by the prevalence of noisy and redundant features, which can lead to overfitting and poor generalisation of the data (Papoutsoglou *et al.*, 2023).

To address these challenges, numerous methods have been developed, including feature selection and extraction. Some methods are specifically designed for microbiome analysis, such as zero-inflated PCA (Zeng *et al.*, 2022). However, it is important to note that different methods can yield varying results on the performance of the predictive model.

### **Importance of Feature Reduction Techniques**

To avoid the curse of dimensionality and the challenges it introduces, it is often necessary to reduce the number of features present in the dataset. Feature reduction is often used to achieve this by retaining only the most informative features while eliminating the irrelevant and redundant features (Popov, 2023). In particular, when handling high-dimensional data from NGS and DNA microarray analysis, feature reduction is essential as many of these features are usually uninformative as they tend to be irrelevant or redundant (Bolón-Canedo, Sánchez-Maróño and Alonso-Betanzos, 2013). Feature reduction can be divided into two categories: feature selection

and feature extraction, which are often considered the same, however, this can be misleading (Sayed Ismail *et al.*, 2024). Feature selection fundamentally involves the selection of relevant features present in the dataset based on some parameter (e.g. correlation, relevancy, etc). Whereas feature extraction involves the creation of a new set of features from the original dataset that contain better representations of the underlying patterns and relationships in the data (Wołczowski and Zdunek, 2017). Principal Component Analysis (PCA) is the most frequently used feature extraction method and is often used to analyse datasets with numerous features or high dimensions.

However, feature selection is used more frequently than feature extraction as it has better interpretability, lower storage, and simpler collection (Zheng *et al.*, 2022). Feature extraction generates new, complex features that can complicate NGS analysis. This is because it becomes challenging to identify the most important sequences; instead of directly indicating sequence significance, the model provides feature importance based on these new, complex features. Feature extraction must still retain all the information of the original features in the dataset thus increasing the storage and slowing downstream tasks. Whereas feature selection only produces a subset of features and does not remember the redundant features (Zheng *et al.*, 2022). As such, it is often recommended to use feature selection over feature extraction for NGS data.

### **Filter-Methods**

Generally, feature selection is divided into three categories: filter, wrapper, and embedded. Filter methods generally rank features based on their score in some statistical tests (e.g. relevancy or correlation) and only features that score above a predefined threshold are selected (Zheng *et al.*, 2022). Filter methods can be divided into univariate methods, which tests each feature individually, and multivariate methods, which tests features as subsets (Pudjihartono *et al.*, 2022). Univariate methods tend to be used over multivariate methods as they have greater speed and simplicity which is an attractive attribute when dealing with high dimensional data. However, univariate methods ignore feature interactions since each feature is tested individually. This results in the inclusion of redundant features in the predictive model, which can decrease its performance (Pudjihartono *et al.*, 2022). The advantage of using filter methods is that it works in isolation from the classifier, hence removing any bias of the learning algorithm from feature selection. As such it concentrates on the overall characteristics of the data, which can reduce overfitting but not fine-tuned to the classifier (Akinola *et al.*, 2022). Maximum-relevance and minimal-redundancy (mRMR) is one of the most commonly used filter feature selection and has

been used in numerous biological and metagenomic papers (Qin *et al.*, 2012, 2012; Elwali and Moussavi, 2021). mRMR uses mutual information to select features that have highest relevance to the class group and are minimally redundant (i.e. maximum dissimilarity with other features). However, Bolón-Canedo, Sánchez-Maróño and Alonso-Betanzos, suggests that other filter methods like ReliefF, and  $M_d$  (an extension of mRMR) provide a better selection of features except in cases of high nonlinearity (Bolón-Canedo, Sánchez-Maróño and Alonso-Betanzos, 2013).

### **Wrapper-Methods**

In contrast, wrapper methods utilise the performance (e.g. classification error, accuracy, etc) of the classifier to evaluate the best feature subset (Bolón-Canedo, Sánchez-Maróño and Alonso-Betanzos, 2013). Wrapper methods have been shown to achieve a higher predictive performance compared to filter methods, however, since wrapper methods need to iterate through the possible feature subsets, training each on a classifier, their computation cost is exceptionally high (Cai *et al.*, 2018). To somewhat alleviate this issue, heuristic approaches are used (e.g. exponential, sequential and, random) to generate the subset of features but their computational cost still far exceeds that of filter methods (Akinola *et al.*, 2022). SVM-RFE (SVM recursive feature elimination) is often evaluated as the optimal wrapper method and is a frequently used method, particularly when handling nonlinear data (Bolón-Canedo, Sánchez-Maróño and Alonso-Betanzos, 2013; Statnikov *et al.*, 2013). However, the Boruta function (wrapper based on a random forest classifier) is frequently used in biological and microbiome classification problems and is more computationally efficient than SVM-RFE, particularly with high-dimensional data (Aagaard *et al.*, 2012; Song *et al.*, 2023; Zhou, Xin and Li, 2023).

### **Embedded-Methods**

Finally, in recent years embedded methods have gained considerable traction due to their high efficiency and performance. Wingfield *et al.*, found that embedded methods perform best out of all feature selection methods when handling metagenomic data. Embedded methods incorporate feature selection within the training process of the classifier by using the internal data from the classifier to determine the optimal features (Akinola *et al.*, 2022). Embedded methods reduce the computational costs compared to wrapper methods as they do not have to retrain the model for each variable subset, however, embedded methods may ignore strongly related features and while selecting poor features reducing the predictive performance of future data (Zheng *et al.*, 2022). LASSO (least absolute shrinkage and selection operator) is the most widely used and often considered the most popular feature selection method due to its simplicity and interpretability

(Hollister *et al.*, 2019; Jiang *et al.*, 2022; Lee *et al.*, 2022). It has been used by numerous researchers in the biological and microbial field, particularly when using a random forest classifier (Marcos-Zambrano *et al.*, 2023). While random forest can also be used as an embedded method by using Gini impurity to determine the optimal features, it is used significantly less frequently compared to LASSO (Loomba *et al.*, 2017; Jiang *et al.*, 2022).

### **Determining N, The Number of Selected Features**

Feature selection methods can also be categorised into two types regarding determining the number of features to be selected: subset evaluation and individual evaluation (Bolón-Canedo, Sánchez-Marño and Alonso-Betanzos, 2013). Subset evaluation produces a candidate group of features as the output, which is determined based on the stabilisation of some performance metric. Whereas individual evaluation produces a ranked list of all features, therefore requiring some threshold to be defined so that less relevant features can be discarded, improving the model performance.

However, there is no established guideline to determine this threshold. Belanche and González, suggested removing features that were two variances away from the mean (Belanche and González, 2011). Meanwhile, Mejia-Lavalle, Sucar and Arroyo-Figueroa, defines the threshold by calculating the gaps between consecutive ranked values from the output and using the largest gap as the threshold (Mejia-Lavalle, Sucar and Arroyo-Figueroa, 2006). However, it is not possible to set a threshold for subset evaluation as they do not produce weights and instead produce a candidate group of features. Therefore, when using a combination of both individual and subset evaluation it may be better and fairer to decide the number of features based on the total number of features as suggested by Bolón-Canedo, Sánchez-Marño and Alonso-Betanzos (Bolón-Canedo, Sánchez-Marño and Alonso-Betanzos, 2013).

However, a recent paper has suggested that performing feature selection in isolation to hyperparameter optimisation may not be the optimal method. Khalil *et al.*, suggests that this method might produce a model with high performance but at the cost of complexity, thereby increasing the interpretability of the model, the likelihood of overfitting, and reducing its generalisation (Khalil *et al.*, 2024). As such Khalil *et al.*, advises the inclusion of feature selection within hyperparameter optimisation, this is still only a suggestion and does not have any theoretical backing to suggest that this is an optimal model (Khalil *et al.*, 2024). Therefore, it may

be advised to include feature selection within hyperparameter tuning to increase the likelihood of selecting relevant features.

## **Hyperparameter and Model Selection for Predictive Analysis**

The selection of an appropriate predictive algorithm is an essential step involved in the analysis of microbiome data through machine learning. There are numerous models that are available for use in regression and classification, each with their own set of strengths and weaknesses (Kuhn, 2019). However, the choice of the predictive model is dependent on the problem type and the characteristics of the data and there is no consensus on a “best” model for any given dataset. When handling microbiome data it is often recommended to select models that are scalable to high-dimensionality and small sample sizes, while remaining interpretable (Papoutsoglou et al., 2023). Therefore, Support Vector Machines (SVMs), Generalised Linear Models (GLMs), Random Forests, and k-Nearest Neighbours (k-NNs) are frequently suggested for use (Bang et al., 2019; Iadanza et al., 2020; Papoutsoglou et al., 2023). Additionally, Gradient Boosted models and Artificial Neural Networks (ANNs) are also used, though less frequently. However, since no single model can universally solve every type of problem, it is often recommended that multiple models be used when performing analysis of the data, and selecting the model that performs best (Zhou, Xin and Li, 2023). Therefore, a careful analysis of the models intended for use, including Generalised Linear Models (GLMs), k-Nearest Neighbours (k-NNs), Support Vector Machines (SVMs), and Random Forests, as well as the corresponding hyperparameters for grid search, has been described in the following section.

### **Generalised Linear Models (GLMs)**

GLMs are some of the most popular and commonly used algorithms for regression and classification. In particular, LASSO and Elastic net regression models are used more frequently than L2 (or ridge) regression (Tay, Narasimhan and Hastie, 2023). GLMs learn a set of weights by minimising their loss function. These are then used to predict the probability of an unknown instance belonging to a specific class. (Sperandei, 2014). The model regularised the coefficients of the model by introducing a L1 (LASSO), L2 (Ridge) penalty or a combination of the two (Elastic net) onto the set of weights. LASSO shrinks many coefficients to zero effectively performing feature selection, whereas Ridge does not shrink any coefficients to zero but does favour solutions with small coefficients. Elastic net incorporates a combination of both LASSO and Ridge by regulating the mixture parameter and determining the strength of regulation between the two model types (Statnikov et al., 2013).

GLMs are consistently used as a benchmark in microbiome analysis as they produce a clear and simplistic model that provides relatively accurate predictions and highly interpretable parameters with low computational efficiency (Pichler et al., 2020). However, GLMs have been known to be prone to underfitting the predictive model, resulting in a lower accuracy compared to other methods. Additionally, normalisation and correlation must be carefully considered when performing GLMs as their performance can be easily affected by inappropriate standardisation methods and high correlation between features (Jiang et al., 2022).

### **K-nearest neighbour (K-NN)**

K-nearest neighbour (K-NN) is a simple machine learning algorithm that classifies an unseen object by a vote of the k nearest neighbours in the training set. In other words, the predicted class of the unseen object is determined to be the class to which the majority of its k-nearest neighbours belong to. The nearest neighbours are calculated using some distance metric, typically Euclidean distance, between the unseen object and all the data points in the training set (Statnikov et al., 2013).

K-NN is generally seen as an efficient and accurate model capable of both classification and regression. However, it becomes significantly more computationally intensive when handling high-dimensional data, which can diminish its efficiency and accuracy. As a result, it often performs similar/worse to GLMs in terms of speed and performance (Pichler et al., 2020; Jiang et al., 2022).

### **Random Forest**

Random forests were originally proposed by Breiman, and have since become one of the most popular classifiers for ML and particularly in microbiome analysis (Breiman, 2001; Pasolli et al., 2016; Chen et al., 2018; Leite et al., 2018). Their popularity is often associated with their inherent ability to analyse high-dimensional and noisy data with high accuracy (Jiang et al., 2022). Random forests are often desired in microbiome analysis due to their ability to handle data with more features than samples, perform embedded feature selection while managing redundant features. They are effective at accounting for correlations between features and can handle both linear and complex interactions (Statnikov et al., 2013). As such, they are versatile in their application and can be used for both binary and multiclass classification of microbiome data.

Random forest is a bagging ensemble of decision trees that adds an extra layer of randomness to increase the diversity of the trees and the general performance of the model. Rather than splitting each node in a decision tree by the best variable, random forest randomly selects a subset of all the variables when creating each tree, resulting in each node being split by the best variable among the subset (Breiman, 2001). The splitting of the node is determined by some criterion (e.g. information gain, Gini impurity, etc).

Each tree within the random forest is used as a base classifier to determine the class of unlabelled instances. These base classifiers or decision trees then cast one vote each for the predicted class label. The class with the most votes is used to determine the label of the unknown instances (Fawagreh, Gaber and Elyan, 2014).

As such random forest is considered an intuitive and interpretable algorithm that is accessible to less experienced users and can prove a powerful predictive tool. However, Topçuoğlu et al., claims that more complex models like random forest are not always necessary to produce the best model. Instead, more interpretable models like GLMs, are capable of producing similar accuracy with significantly less computational complexity (Topçuoğlu et al., 2020).

### **Support Vector Machines (SVM)**

SVMs are powerful algorithms introduced by Vladimir Vapnik in the 1990s as a linear classification approach for pattern classification, using the theory of kernel-induced spaces (Eitrich and Lang, 2006). If the dataset is not **considered** linearly separable, a nonlinear kernel/function can be used to map the data to a high dimensional feature space. SVMs have since grown to become one of the most successful and widely used classifiers and many developments have been made in recent years that aim to improve its performance (Wang, Dong and Hua, 2020).

The basic idea of a SVM is to separate two classes of instances by constructing two hyperplane boundaries that separate two classes of instances, while maximising the distance between the hyperplanes. fitting the instances into the feature space and ensuring a generalised model (Wang and Meng, 2011). Many researchers have used SVMs to analyse NGS data as they can often manage many of the challenges associated with this type of data (Leite et al., 2018; Bang et al., 2019; Su et al., 2022). SVMs are capable of dealing with high ratios of variables to samples and are robust to large numbers of features. Additionally, they can learn complex patterns and use regularisation techniques to prevent overfitting (Statnikov et al., 2013).

## **Hyperparameter Optimisation**

After model selection, the next task is a complex and time-consuming process involving the development of an optimal model architecture via hyperparameter tuning. Hyperparameters are parameters that cannot be directly estimated from data learning and must be set prior to training a model (Yang and Shami, 2020). As such, hyperparameter tuning involves optimising the choice of the parameters that need to be set before training the model, by exploring a range of possibilities. Various methods are available for hyperparameter optimisation, with grid search being one of the most commonly used (Rasool et al., 2022). Grid search employs cross-validation and a brute-force approach to find the optimal parameter values by evaluating all possible combinations within a predefined grid and selecting those that achieve the highest cross-validation metric score.

It's essential to recognise that hyperparameter optimisation is inherently model-specific, each predictive model has its own unique set of hyperparameters that can significantly impact its performance. Consequently, this variation makes the process of sectioning the optimal model and determining the appropriate range of parameters for tuning challenging. Although there is no consensus for a set of optimal hyperparameters or definitive range of values to choose from, guidelines can often be found within literature and can serve as a necessary starting point (Breiman, 2001; Hassanat et al., 2014). However, it is important to clarify that these recommendations are not guaranteed to determine the optimal values and the effectiveness of suggested hyperparameter ranges can vary depending on the specific dataset and problem at hand. Therefore, some form of trial and error is usually necessary.

## **Cross-Validation For Model Improvement**

Cross-validation (CV) has been and continues to be widely used in literature to determine model selection (Carracedo-Reboredo et al., 2021). CV involves a process of splitting the data into subsets and using them to iteratively train and test the predictive performance of candidate models (Albaradei et al., 2021). The general goal of CV is to determine the optimal parameter combinations based on the hyperparameter selections, thus producing an optimal model (Carracedo-Reboredo et al., 2021). The simplest form of cross-validation is known as leave-one-out, which uses all data points except one to create a candidate model. The remaining data point is used to test the predictive performance of the candidate model (Poldrack, Huckins and Varoquaux, 2020). Leave-one-out is often problematic as the test set is not representative of the

whole dataset and can sometimes introduce correlation biases. Therefore, leave-one-out should only be used in the most extreme cases when sample size is significantly deficient. Instead, an alternative is to use leave-k-out, with k being 10-20% of the data or to use K-fold CV (Poldrack, Huckins and Varoquaux, 2020).

K-fold CV is a more general approach that involves randomly and evenly splitting the data into K subsets (or folds). The model is then trained on K-1 folds, known as the training set, with the remaining fold being held as the validation set (Jung and Hu, 2015). The validation set is used to evaluate the candidate model, providing a CV score (typically some performance metric). Each possible combination of K-1 folds is used to create a candidate model, and the optimal model is selected from all the candidate models based on the “best” CV score (e.g. highest accuracy) (Jung and Hu, 2015).

There is no consensus on the optimal value of K, but most studies evaluating its effectiveness observe a decrease or constant in variance and bias as K increases (Kohavi, 1995; Bengio and Grandvalet, 2004). For datasets with smaller sample sizes, variance and bias tend to stabilise when K approaches 10, while this convergence occurs sooner with larger sample sizes. However, some studies have reported an increase in variance with higher K values, primarily in LASSO and SCAD models (Zhang and Yang, 2015).

While K-fold cross-validation is the most widely used method, other cross-validation techniques can offer better performance for specific situations (Molinaro, Simon and Pfeiffer, 2005). Notable alternatives include: (1) Repeated CV, which involves refitting the predictive model multiple times (typically 10) to further reduce variability, (2) Stratified CV, which ensures that each fold maintains the same proportion of class distributions as the entire dataset, and (3) Nested CV, which performs an additional layer of cross-validation to optimise hyperparameters and prevent overfitting (Krstajic et al., 2014). Each of these methods can provide distinct advantages depending on the problem at hand. Repeated CV tends to be used as an alternative for K-fold CV when the sample size is small as it reduces the variation.

Particularly when handling high-dimensional data relative to sample size (like NGS data), Wilimitis and Walsh encourages the use of nested CV over other methods as it provides a simple way of performing both model selection and hyperparameter tuning while mitigating further bias Wilimitis and Walsh, 2023. Similarly to K-fold CV, nested CV does not have any consensus regarding the

number of folds for the outer or inner loop. Additionally, these are not required to be the same or different value (Krstajic et al., 2014).

## **Interpreting Biological Significance Through Feature Importance in Data Analysis**

In general, there exists two main goals of performing statistical modelling of data: either the generation of a model or extracting features of interest. However, some domains require both predictive models that are capable of providing an accurate forecast of the data and interpretable results (Altmann *et al.*, 2010). Interpretable results are those that provide some information about the features and how they relate to the predictions of the model, while being comprehensible and understood by the human mind (Marcinkevičs and Vogt, 2023). Explained and interpretable models are in greater demand in the medical and biological domain as it is important for models to predict the outcome of the patient but also provide actionable advice for prevention or insight into the underlying causes of the diseases (Saarela and Jauhiainen, 2021). However, Tjoa and Guan state that the number of medical and biological papers that address the issues of explainability and interpretability are limited (Tjoa and Guan, 2021). Therefore, it is necessary to prioritise the inclusion of explainable models in these fields. Linear models are typically used as they prove easier to interpret than nonlinear models. However, linear models usually perform worse than their nonlinear counterparts (Musolf *et al.*, 2022). As a result, it's important to find ways to make nonlinear models more interpretable.

### **Global vs Local Feature Importance**

As such, feature importance analysis is a crucial step involved in gaining insight into how these complex algorithms work. Feature importance estimates the individual contribution of each feature using a particular score, ranking or metric, to the classifier output. Using feature importance helps to define the causality between important features and the outcomes of the predictive model by clarifying how the model arrived at its decisions (Rengasamy *et al.*, 2022). Feature importance can be characterised by different properties. First, it can be categorised into either Local or Global importance. Local importance measures the contribution of a feature on the classification performance for a specific instance. Whereas Global importance provides insight into the entire model by measuring the importance of a feature on the classification performance for the entire data (Alfeo *et al.*, 2023). In general, Global importance is more frequently used than Local importance as there is often no need to investigate specific instances. However, in the field of

personalised medicine, Local importance is crucial as it can aid anaesthesiologists in tailoring treatments and identifying the specific causes of illness in individual patients (Adler and Painsky, 2022).

### **Agnostic vs Specific Models**

Second, there can be either model-agnostic or model-specific evaluation methods for feature importance. Model-agnostic methods generate results independent of the trained model and can be applied in the same way for any classifier. Whereas model-specific methods rely on the trained model for their results, therefore they can only be used with specific classifiers (Adler and Painsky, 2022). The most frequently used local model-agnostic method is either SHAP (SHapley Additive exPlanations) or LIME (Local Interpretable Model-agnostic Explanations) (Muschalik *et al.*, 2022). Model-specific methods are not designed for local feature importance but are commonly used for global feature importance. Some of the more frequently used methods include logistic regression which uses the coefficients produced from the model to rank the importance of each feature and random forest which uses Gini impurity to also rank the features. However, perhaps the most used approach for global feature importance is permutation importance (PIMP), a model-agnostic method (Adler and Painsky, 2022). PIMP is often preferred over model-specific methods as it can be used across multiple classifiers, indicates significance of each feature (i.e. p-value) and does not introduce bias in favour of features with more categories (Altmann *et al.*, 2010).

### **Ensemble Feature Importance**

Using different models or different feature importance methods can generate different feature importance values which can reduce the reliability of results and their interpretations. Thus, there have been efforts made to improve the accuracy of feature importance methods. One such method was introduced by Rengasamy, Rothwell and Figueredo, who suggested the use of an ensemble feature importance method (Rengasamy, Rothwell and Figueredo, 2021). This applied multiple feature importance methods to a set of ML models and combined their results to produce a final importance score that reduced the overall error score by 15%. Originally ensemble methods involved using Gini impurity from multiple decision trees to calculate feature importance but have since improved to gather feature importance from either multiple feature importance methods or ML classifiers (Breiman, 2001).

## Summary

This study focuses on utilising machine learning techniques to analyse the human gut microbiome, with an emphasis on their impact on the athletic gut microbiome. Traditional methods, such as PCoA, differential abundance, and diversity analysis, often fall short in handling the complexity and high dimensionality of microbiome data, which involves numerous microbial species and their interactions. Machine learning offers a powerful alternative by providing advanced tools for classification and pattern recognition that can uncover subtle relationships between gut microbiota and athletic performance.

The literature review covers critical aspects of applying machine learning to microbiome data, including preprocessing techniques necessary to prepare data for analysis. This includes challenges like rarefaction, compositionality, and imputation that are specific to microbiome and NGS studies. It also addresses the importance of feature reduction techniques to manage high-dimensional data and discusses various machine learning algorithms and their tuning for optimal performance. Additionally, the review examines how to interpret the biological significance of features identified by these algorithms, a crucial step in translating machine learning findings into actionable insights about how specific bacterial species may influence athletic outcomes.

# Methodology

## Data Usage

This study utilised human gut microbiome data derived from two previously conducted investigations on the athletic microbiome (Scheiman et al., 2019; Fontana et al., 2023). Scheiman et al., involved marathon runners and control participants, with faecal samples being collected from both groups daily for up to one week before and one week after the marathon. The faecal samples were used to characterise the bacterial composition of each participant's gut microbiome using 16S rDNA amplicon sequencing. The sequenced data was processed to generate a sequence table with taxonomy assigned across all possible taxonomic levels. Furthermore, the abundance was calculated at each level, resulting in a count table that lists the samples and their corresponding abundance for every taxonomic feature at all levels.

For my analysis, I extracted the count data from this study, specifically focusing on the genus, species, and RSV levels of the identified bacterial taxa. This data served as the input for the machine learning framework developed in this thesis. Genus-level taxonomic resolution was used to explore broad ecological insights into microbial communities but lacks the detailed microbial differentiation that species or RSV-level analysis would offer. As a result, it is necessary to utilise all three taxonomic levels.

Fontana et al. utilised publicly available shotgun metagenomic data from athletes, ensuring consistency by using the same sequencing technology across 185 samples from high anaerobic and aerobic load sports (e.g., cycling, rugby), 164 samples from healthy sedentary controls, and 69 samples from moderate athletes, all sourced from the NCBI repository. Unlike Scheiman et al., who provided raw abundance data, Fontana et al. utilised relative abundance data, specifically at the species and EC (Enzyme Commission) levels. Since EC codes are less interpretable and less relevant for linking microbiota to the athletic GM, this analysis focused exclusively on species-level data, resulting in slight variations in the inputs used for the machine learning framework.

## Experimental Design and Parameter Grid

This study proposes a machine learning framework for effective application of microbiome-based classification problems, specifically using it to analyse the gut microbiomes of athletes as an example. The framework offers a reproducible pipeline for training, evaluating, and interpreting microbiome-based machine learning models, while also generating hypotheses to explain the

underlying biological features driving the model predictions (Figure 1). By adhering to best practices outlined in recent publications, the framework ensures the accurate and reasonable selection of methods within the model. The framework consists of five main steps: preprocessing, feature selection, hyperparameter tuning and model selection, performance evaluation and plotting, and feature importance analysis, utilising the caret package (v.6.0.94) in R (v.4.3.3) and repeated CV. Additionally, it requires a sequence table, associated metadata, and specific parameters that guide how each step is performed.

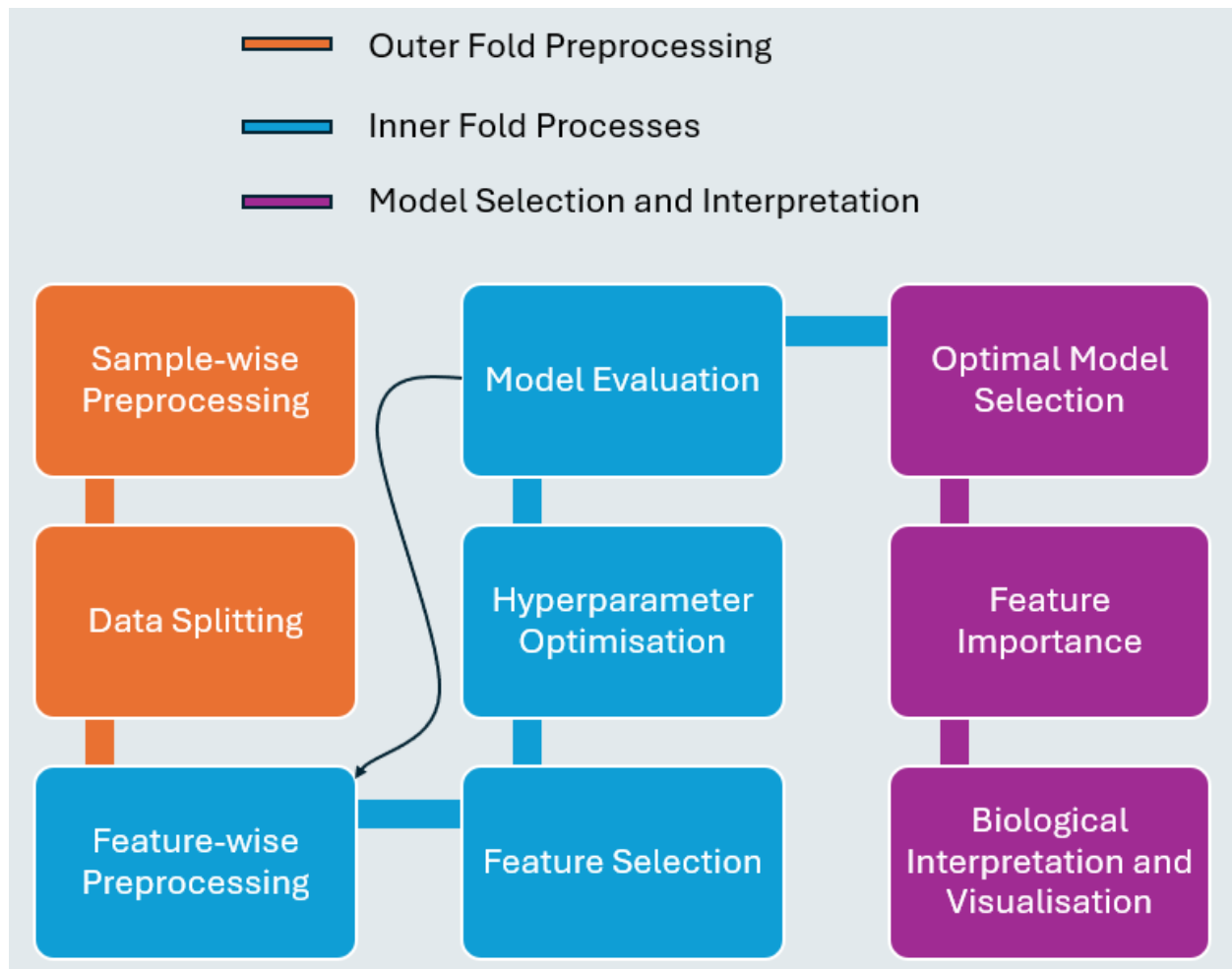


Figure 1: Key steps involved in the machine learning pipeline. The pipeline is divided into three stages: (1) preprocessing prior to data splitting, (2) processes carried out within each outer fold (each process is repeated for each outer fold), and (3) model interpretation, which occurs after all models have been evaluated across the outer folds.

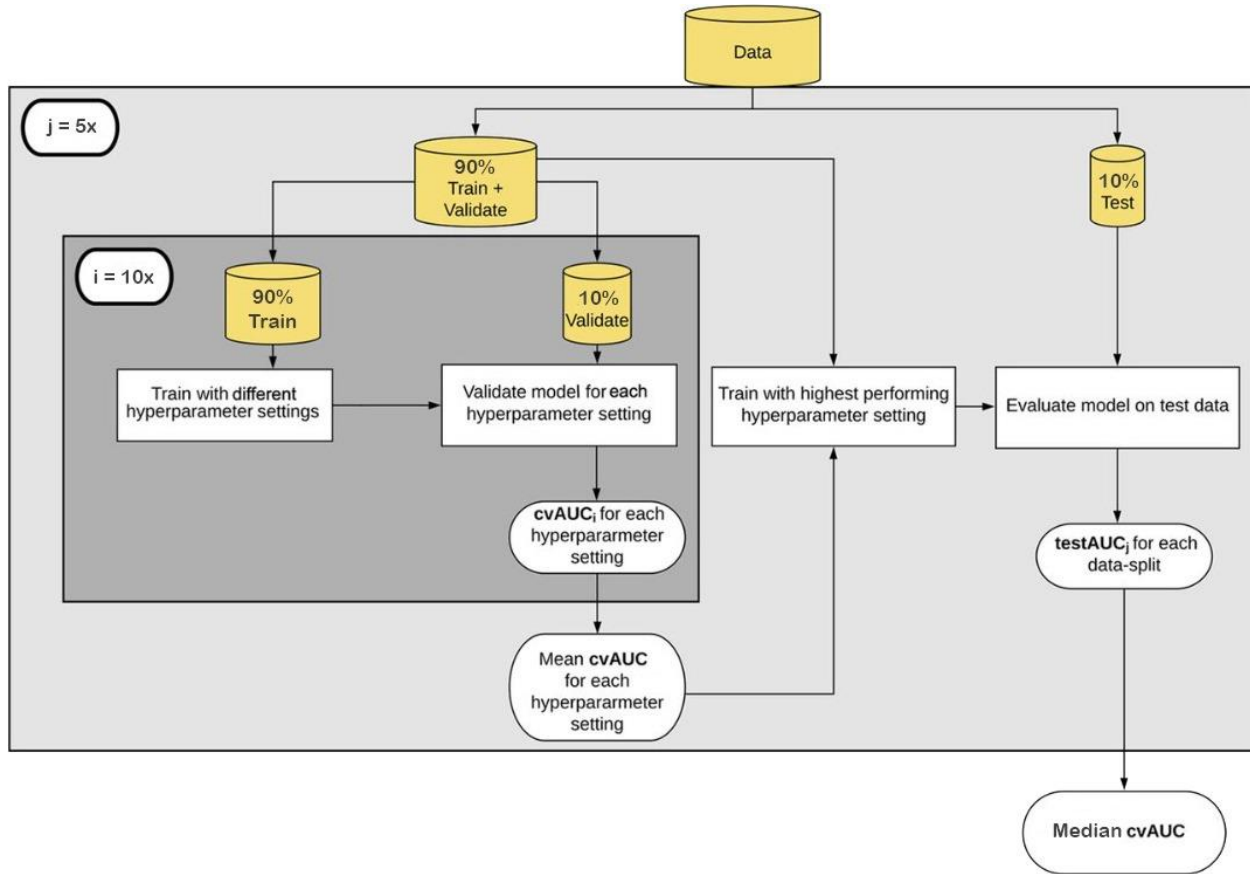


Figure 2: Machine learning pipeline. The data was split to create a training (90%) and held-out test set (10%). The splits were grouped according to subject ID to prevent data leakage. A ten-fold cross-validation was performed on the training data to select the best hyperparameter setting and then used these hyperparameters to train the models. The model was evaluated on the held-out data set. Abbreviations: cvAUC, cross-validation area under the curve. Adapted from Topçuoğlu et al., 2020.

To identify the optimal machine learning models for classification, a comprehensive approach was taken, systematically testing a wide range of model configurations. This process involved creating a structured parameter grid to explore all possible combinations of key parameters across all steps included in the framework. The parameter grid was then used to systematically produce a distinct model configuration for each combination of parameters, followed by training and validation using repeated cross-validation (CV). The performance of each model was then evaluated based on the relevant performance metric (e.g., AUC for classification).

By systematically assessing each combination of the key parameters, 288 distinct models were produced for each taxonomic level using the 16S rDNA sequencing data. An additional 108 distinct models were produced from the shotgun metagenomic species level data. There is a reduction in the total number of models as the format of the input data is a sequence table with relative abundance data as opposed to raw abundance data. This renders some of the initial preprocessing parameters redundant or incompatible with the data, leading to a lower overall model count.

Once evaluated, the models were ranked and ordered based on their performance metrics. This ranking process allowed for the identification of the best performing models, which were selected for further analysis and biological interpretation. By employing this rigorous and exhaustive approach to model selection and evaluation, the research ensured that the most effective machine learning models were identified, contributing to the robustness and reliability of the outcomes.

## **Preprocessing**

The pipeline begins with preprocessing samples which is specific to handling NGS and microbiome data. Here, the zero values within the count matrix are imputed using either a pseudo count (1 or  $0.65 \times \min$ ) or Bayesian Multiplicative Replacement (CZM method from the zCompositions R package). Following this, the count matrix undergoes transformation through either Centred Log-Ratio (CLR), robust CLR (rCLR), or Total Sum Scaling (TSS). CLR and rCLR are employed to address the constraints of compositional data, with rCLR serving as an adaptive alternative to CLR, whereas TSS is used to maintain the compositional nature of the data, avoiding imputation or offer a simple, intuitive approach.

Data splitting involves splitting the data into k-folds for the outer partition of the pipeline. To ensure reliability and prevent data leakage, all samples from the same unique group are kept within either the training or testing set during each fold and the class weights are assigned to address data imbalance. Each fold may have a varying number of samples between training and test sets depending on how the unique groups are assigned. Sample-based preprocessing is kept outside of data splitting because imputation and scaling are unaffected by the fold splits and performing them separately within each fold would unnecessarily increase computational demand. However, it is crucial that feature-based preprocessing is conducted separately within each fold to prevent data leakage and ensure valid model evaluation.

Feature-based preprocessing can utilise numerous methods to process the input data. For this study, redundant features were removed using near-zero variance filtering (NZV), and the data was z-score standardised, reducing the mean to 0 and the standard deviation to 1. Additionally, perfectly correlated features are removed based on a stringent Spearman test (100% correlation), reducing the burden on downstream feature selection methods and eliminating obviously redundant features.

## **Feature Selection**

For feature selection, six different approaches were initially considered: LASSO, Boruta, SVM-RFE, mRMR, T-test, and Wilcoxon test. However, only three methods were ultimately chosen for model testing. mRMR, being a multivariate method, was selected as a more robust alternative to the simpler univariate T-test and Wilcoxon test, rendering them redundant. mRMR selects a predefined number of features (N) for inclusion in the model and operates with default settings. Similarly, Boruta selects a predefined N while also using default settings. LASSO uses the same hyperparameters and cross-validation method as employed in model selection (discussed later), with the exception of alpha, which is specifically set to 0 to create a true LASSO model. LASSO also selects a predefined N. Although SVM-RFE was initially considered for inclusion, it was ultimately excluded due to its incompatibility with top N feature selection and its significantly higher computational demands compared to the other methods.

## **Hyperparameters Optimisation**

To determine the relationship between microbiota and the athlete microbiota, four machine learning modelling techniques were explored: GLM, SVM (with radial basis), k-NN, and RF. The goal was to identify their optimal configuration parameters and determine which models yielded the best performance scores. Therefore, model selection requires hyperparameter tuning, which is achieved by performing a ten-fold repeated CV on the training data of each outer-fold. This creates ten inner-folds for each outer-fold, specifically designed for hyperparameter optimisation and repeats the validation ten times. The ten-fold repeated CV ensures that all samples of each unique group are contained in either the training or validation data, not both. The optimal hyperparameter values were based on the best average CV predictive performance using the area under the curve (AUC).

### **GLM**

The two hyperparameters involved in GLM modelling are alpha and lambda. Alpha regulates the mixture parameter between LASSO (0) and Ridge (1). Selecting a value between 0 and 1 determines the strength of regulation with 0.5 being a balance between the two model types (Buczak et al., 2024). Alpha is often considered a higher-level parameter where its value depends on the type of prediction model desired (e.g. LASSO, Ridge, or Elastic net). Therefore, it is often unnecessary to include within the tuning grid (Tay, Narasimhan and Hastie, 2023). However, a sequence of five values ranging from 0-1 were tuned for alpha as it can be beneficial to thoroughly identifying the optimal model.

Lambda indicates the degree of penalisation to the coefficients of the model. For tuning of lambda, it is often suggested to create a sequence of 100 values from the minimum lambda to the maximum lambda equally spaced along a log scale (An Introduction to `glmnet`, 2023). The minimum lambda is set as  $10^{-4} \times \text{maximum lambda}$ . Maximum lambda is equal to the smallest value of lambda that would drive all coefficients to zero. The default maximum lambda was set to 10 as the coefficients of sample models were driven to zero by then. However, this may not be the optimal value for all the models.

## **K-NN**

The most crucial aspect of K-NN is determining the optimal number of nearest neighbours  $k$ . The classification performance is highly dependent on the value of  $k$ . If  $k$  is not sufficiently large the model will underfit, reducing the reliability of the model results, whereas if  $k$  is too large the model will be overfitting, impairing the model's ability to generalise (Yang and Shami, 2020). However, selection of  $k$  can be difficult as the optimal value differs between datasets. Therefore, it is often necessary to perform hyperparameter tuning of  $k$  to determine the optimal value, using a range of possible values for  $k$ .

The range often starts at a minimal value of  $k = 1$  and ends with a maximum value of number of sample (Jirina and Jirina, 2010; Hassanat et al., 2014). This is a general rule of thumb that is often used as a starting point in determining the optimal value of  $k$ , but further refinement necessary to determine the precise optimal value. As such, a sequence of ten values ranging from 1 to number of samples were used for  $k$  to reduce the computational demand of the optimisation. Even numbers for  $k$  were avoided as well to increase computational speed and to avoid uncertainty which can be introduced into the model if two classes have the same number of votes (Hassanat et al., 2014).

## Random Forest

The hyperparameter Mtry is often considered one of the most important aspects of implementing randomness into the classifier as it determines the number of features sampled when splitting a node. A lower Mtry value improves randomness and reduces variance but may reduce the accuracy if it is too low, whereas increasing Mtry increases likelihood of overfitting but can increase accuracy (Liaw and Wiener, 2001). The default value for classification is often considered number of features. However, this study follows Breiman, 2001 suggestion for hyperparameter tuning and using half of the default, the default and twice the default as the initial starting point.

## SVM

The quality of an SVM model is highly dependent on the selection of the hyperparameters within the model. Therefore, the main issue for practitioners applying an SVM model is selecting the correct set of hyperparameters (Cherkassky and Ma, 2004). However, there is no widely agreed upon optimal parameter. As such, hyperparameter tuning remains the method of choice when determining their values.

Parameter C determines the trade-off between generalisation of the data by maximising the margin of the hyperplane and accurately classifying the training data by controlling the error toleration (Eitrich and Lang, 2006). A larger C value selects a smaller margin favouring classification of the training data over generalisation, whereas a smaller C value increases the margin, favouring generalisation. As such, a range of values along a logarithmic scale are used as the starting point for hyperparameter tuning. SciKit-learn recommends a range of  $10^{-3}$  to  $10^3$ , which is sufficient to find a relatively optimal value (RBF SVM parameters, 2024).

Parameter sigma determines how the decision boundary of the hyperplane is formed. Sigma affects both the reach and number of support vectors that are used to construct the boundary (Cherkassky and Ma, 2004). A larger value will only select support vectors closest to the decision boundary, resulting in fewer support vectors selected. A smaller value will consider support vectors further from the decision boundary, selecting more points overall. SciKit-learn similarly recommends a range of  $10^{-3}$  to  $10^3$  for sigma as well (RBF SVM parameters, 2024). For this study, the recommendations by SciKit-learn were used for C but not Sigma as a smaller range of values ( $10^{-6}$  to  $10^{-1}$ ) were found to be more favourable when tested on the sample models.

## **Model Evaluation and Interpretation**

After selecting the hyperparameters, the model was trained using the training dataset from each outer fold. The trained model from each fold was then applied to its respective held-out data to evaluate testing predictive performance. This process generated a performance table, an ROC plot, and a confusion matrix to visualise model performance. Each outer fold produced a final model, along with its corresponding performance table and plots. The AUC performance metric was extracted from each final model, and the model with the median AUC was selected as the representative final model for the framework.

Finally, feature importance analysis was performed using a permutation importance (PIMP) approach. PIMP, being model-agnostic, ensured that feature importance results across the model collection were consistent in format, allowing for straightforward and more robust interpretation. Although model-specific approaches are available within the pipeline, they were not used during the testing of the model collections.

## **Biological Interpretation and Visualisation**

This pipeline was repeated for each combination of key parameters within the parameter grid, resulting in a total of either 288 or 108 models. The models were then ranked and ordered according to their performance. After model ranking, the analysis focuses on identifying the features that were the most significant for differentiating athletes from sedentary controls, evaluating the effectiveness of different models, and determining the parameters that produced the optimal performance.

To achieve biological interpretation and visualisation of the machine learning results, all the biological features selected by the final models were extracted, and their percentage coverage across the entire model collection was calculated. These were categorised into distinct percentage groups and their distribution plotted. Utilising the feature importance from the model collections, statistically significant ( $p\text{-value} \leq 0.05$ ) features were extracted and both their total frequency and significance frequency across the entire model collection was plotted to visualise their relevance. Lastly, based on these plots, the most significant features and their relative abundance data were used to highlight the difference in microbial composition between the control and athlete groups.

## **Additional Notes**

This study followed guidelines from Marcos-Zambrano et al., to avoid common problems associated with machine learning analysis.

The code for all sequence table analysis and visualisation steps is available at <https://github.com/studentUCC/SeqBiomeML.git>

For all steps, a seed value of 0 was used, except at the genus taxonomic level, where a seed of 100 was applied. Using seed 0 at the genus level caused the outer fold splitting to result in a test set that included only control groups and no athlete groups, leading to issues with performance calculations.

## Results

The study repeats the pipeline for each combination of key parameters within the parameter grid, resulting in a total of either 288 (for the 16S rDNA data) or 108 (for the metagenomic data) models. The models were then ranked and ordered according to their performance. After model ranking, the analysis focuses on identifying the features that were the most significant for differentiating athletes from sedentary controls, evaluating the effectiveness of different models, and determining the parameters that produced the optimal performance.

### Model Interpretation

The effectiveness of the models and their optimal parameters were evaluated based on the median AUC value for each modelling approach and their variance in performance across the entire model collection. Due to time constraints and coding challenges, only ten key parameters were properly analysed (Table 1). On average, the AUC range within each of these parameters (e.g. Random Forest mode or mRMR feature selection) varied by 0.281 across all the models and datasets. If only a single value had been examined for each specific combination, the insights gained would have been limited. The model's performance could have been influenced by chance or the particular method or model type chosen, rather than providing a reliable assessment of its true effectiveness. For instance, the lowest AUC value of the random forest model was 0.691, whereas the highest was 0.880.

| <b>Model Parameters</b> | <b>Min.</b> | <b>1st Qu.</b> | <b>Median</b> | <b>Mean</b> | <b>3rd Qu.</b> | <b>Max.</b> |
|-------------------------|-------------|----------------|---------------|-------------|----------------|-------------|
| CLR                     | 0.631       | 0.768          | 0.803         | 0.796       | 0.841          | 0.917       |
| rCLR                    | 0.595       | 0.730          | 0.776         | 0.769       | 0.820          | 0.888       |
| TSS                     | 0.544       | 0.732          | 0.783         | 0.765       | 0.817          | 0.923       |
| Boruta Filter           | 0.629       | 0.778          | 0.814         | 0.803       | 0.842          | 0.920       |
| GLM Filter              | 0.623       | 0.760          | 0.803         | 0.788       | 0.825          | 0.917       |
| mRMR Filter             | 0.608       | 0.753          | 0.796         | 0.784       | 0.824          | 0.887       |
| Glmnet                  | 0.664       | 0.777          | 0.806         | 0.802       | 0.840          | 0.895       |
| kkNN                    | 0.587       | 0.712          | 0.754         | 0.752       | 0.806          | 0.900       |
| RF                      | 0.691       | 0.778          | 0.817         | 0.808       | 0.840          | 0.915       |
| svmRadial               | 0.658       | 0.773          | 0.812         | 0.804       | 0.847          | 0.880       |

Table 1: Summary of Mean Statistics for Key Parameters Across All Taxonomic Levels Based on AUC

However, the median predictive performance of the random forest model was higher than other ML models with a value of 0.817 (interquartile range [IQR], 0.778 to 0.840), though not significantly ( $P = 1$  [The P-value was calculated using Dunn's post-hoc test to identify pairwise differences between the groups, following a significant result from the Kruskal-Wallis test]) (Table 2). These results showed that depending on the combination of parameters, the testing performance can vary but the median value is generally higher than other models (Figure 3). Therefore, it is important to employ multiple sample points when estimating generalisation performance.

| Comparison of Model Parameters | Z-score | P.unadj  | P.adj           |
|--------------------------------|---------|----------|-----------------|
| CLR - rCLR                     | 2.331   | 1.98E-02 | 5.93E-02        |
| CLR - TSS                      | 3.563   | 3.66E-04 | <b>1.10E-03</b> |
| rCLR - TSS                     | 0.219   | 8.27E-01 | 1.00E+00        |
| Boruta Filter - GLM Filter     | 1.486   | 1.37E-01 | 4.12E-01        |
| Boruta Filter - mRMR Filter    | 3.503   | 4.59E-04 | <b>1.38E-03</b> |
| GLM Filter - mRMR Filter       | 2.017   | 4.37E-02 | 1.31E-01        |
| Glmnet - kkNN                  | 6.646   | 3.01E-11 | <b>1.80E-10</b> |
| Glmnet - RF                    | -1.071  | 2.84E-01 | 1.00E+00        |
| kkNN - RF                      | -7.718  | 1.19E-14 | <b>7.11E-14</b> |
| Glmnet - svmRadial             | -0.597  | 5.50E-01 | 1.00E+00        |
| kkNN - svmRadial               | -7.243  | 4.38E-13 | <b>2.63E-12</b> |
| RF - svmRadial                 | 0.474   | 6.35E-01 | 1.00E+00        |

Table 2: Summary of Dunn Test Results for Key Parameters Across All Taxonomic Levels

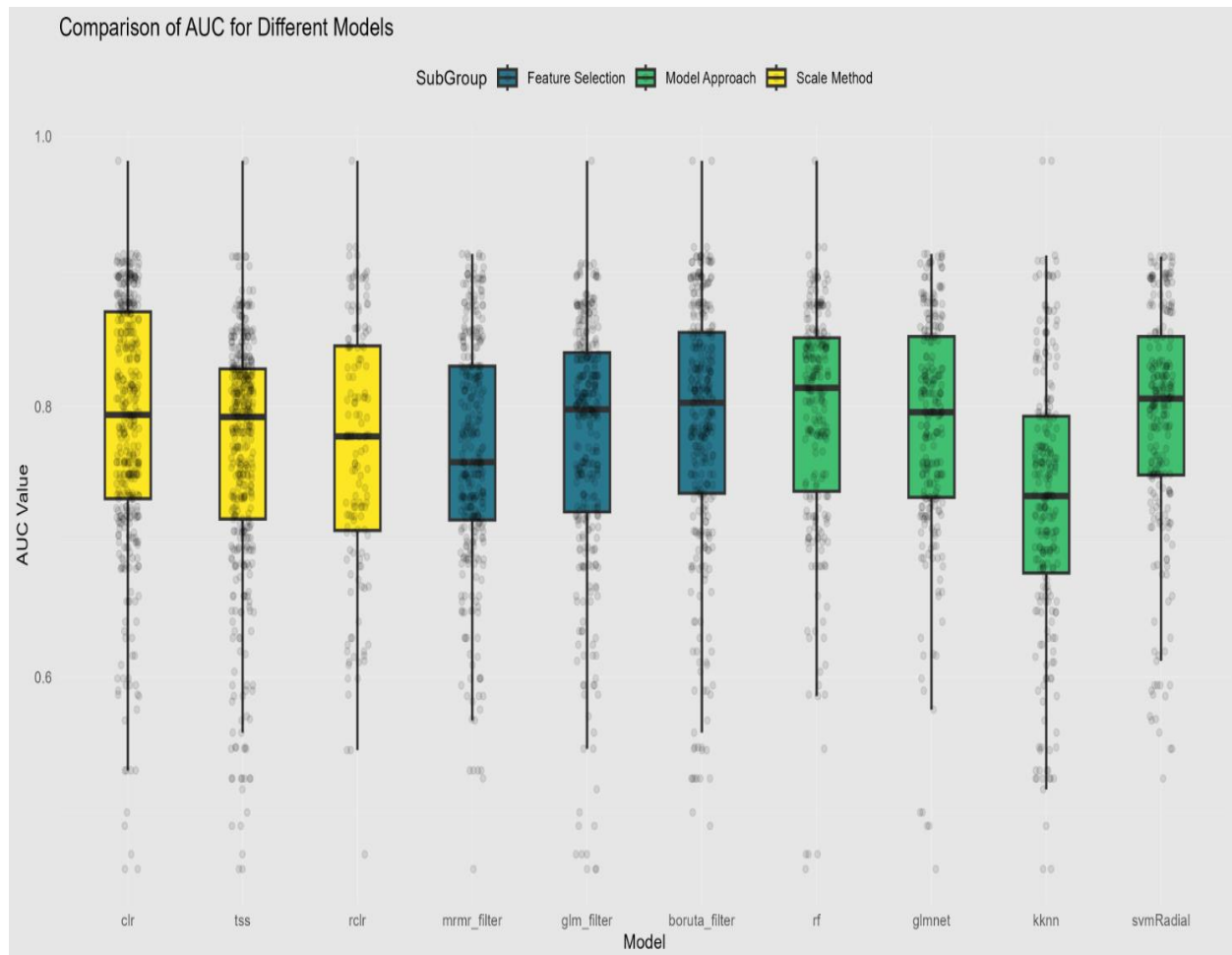


Figure 3: Comparison of AUC for different machine learning models. The boxplot displays the AUC values for various models, categorised by subgroups: Scale Method, Feature Selection, and Model Approach. The median AUC values for each model are shown as horizontal lines within the boxes, with quartiles represented by the box ends. Whiskers extend to the farthest data points that are not outliers. Models are compared across different approaches, the colours representing the different groups. Each point represents an AUC value from individual model.

Similarly, GLM and SVM with radial basis AUC values were not significantly different from one another and had median AUC values of 0.803 (IQR, 0.760 to 0.825), and 0.812 (IQR, 0.773 to 0.847), respectively. K-NN had significantly ( $P = 7.11\text{e-}14$ ,  $1.80\text{e-}10$ ,  $2.63\text{e-}12$ , respectively) lower AUC values than the other ML models with median AUC of 0.754 (IQR, 0.712 to 0.806) (Figure 3). These results demonstrate that the models typically better suited for scaling with larger datasets (GLM, SVM with a radial basis, and Random Forest), outperformed K-NN, which may struggle with scalability and efficiency in larger or more complex datasets.

The highest median predictive performance of the feature selection methods was the Boruta method with a value of 0.814 (IQR, 0.778 to 0.842), however, not significantly different from the GLM feature selection method ( $P = 0.412$ ), which had a median value of 0.803 (IQR, 0.760 to 0.825). mRMR had a significantly ( $P = 0.001$ ) lower median AUC value of 0.796 (IQR, 0.753 to 0.824) compared to the Boruta feature selection method (Figure 3). Notably, no significant difference was observed between wrapper and embedded methods, although the filter method performed worse in comparison.

Regarding sample preprocessing methods, CLR had the highest median predictive performance with a value of 0.803 (IQR, 0.768 to 0.841) though not significantly different from rCLR ( $P = 0.059$ ), which had a value of 0.776 (IQR, 0.730 to 0.820). TSS had a significantly ( $P = 0.001$ ) lower median AUC value of 0.783 (IQR, 0.732 to 0.817) compared to the CLR preprocessing method (Figure 3). Unsurprisingly, no significant difference was found between CLR and rCLR considering they are similar techniques and that TSS underperformed compared to CLR.

## **Biological Interpretation**

To achieve biological interpretation and visualisation of the machine learning results, all the biological features selected by the final models were extracted, and their percentage coverage across the entire model collection was calculated (Figure 4). The distribution of feature coverage followed an exponential pattern, with most features appearing in less than 25% of the models, while a few occurred in over 95% of the models. Notably, no feature at any taxonomic level was present in every model. This suggests that although some features are highly effective in distinguishing between control and athlete groups, no single feature can perfectly explain the differences in athletic performance. When comparing the most frequent compositions of microbial species between the 16S rDNA data and the metagenomic data, it was found that the spectra of

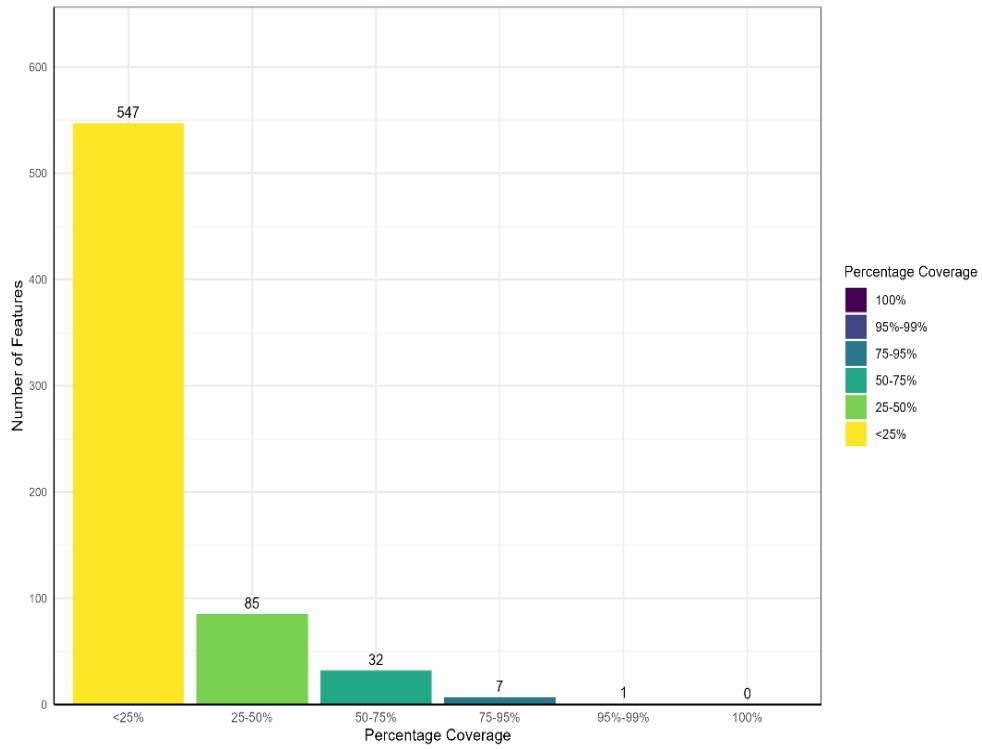
microbial species mostly differed. In the 16S rDNA data, the most dominant microbial species were *Phascolarctobacterium faecium* (276), *Bacteroides cellulosilyticus* (273), *Parasutterella excrementihominis* (256), *Bacteroides vulgatus* (251), *Faecalibacterium prausnitzii* (236); however, in the metagenomic data, the most dominant species were *Bacteroides faecis* (96), *Phocaeicola vulgatus* (96), *Bacteroides caccae* (92), *Alistipes communis* (89), *Ruminococcus unknown\_species* (88) (Table 3).

| Most Frequently Occurring Features            | Frequency of Occurrence | Most Frequently Significant Features        | Frequency of Significance |
|-----------------------------------------------|-------------------------|---------------------------------------------|---------------------------|
| <b>RSV-Level</b>                              | <b>Max = 288</b>        | <b>RSV-Level</b>                            | <b>Max = 288</b>          |
| Seq_0000070# Phascolarctobacterium faecium    | 282                     | Seq_0000070# Phascolarctobacterium faecium  | 168                       |
| Seq_0000093# NK4A214 group                    | 266                     | Seq_0000093# NK4A214 group                  | 158                       |
| Seq_0000068# Parasutterella excrementihominis | 260                     | Seq_0000001# Bacteroides vulgatus           | 101                       |
| Seq_0000001# Bacteroides vulgatus             | 253                     | Seq_0000432# [Eubacterium] ventriosum group | 93                        |
| Seq_0000432# [Eubacterium] ventriosum group   | 245                     | Seq_0000020# Bacteroides cellulosilyticus   | 91                        |
| <b>Species-Level</b>                          |                         | <b>Species-Level</b>                        |                           |
| Phascolarctobacterium faecium                 | 276                     | Bacteroides cellulosilyticus                | 139                       |
| Bacteroides cellulosilyticus                  | 273                     | Parasutterella excrementihominis            | 134                       |
| Parasutterella excrementihominis              | 256                     | Phascolarctobacterium faecium               | 126                       |
| Bacteroides vulgatus                          | 251                     | Sutterella wadsworthensis                   | 81                        |
| Faecalibacterium prausnitzii                  | 236                     | Bacteroides vulgatus                        | 70                        |
| <b>Genus-Level</b>                            |                         | <b>Genus-Level</b>                          |                           |
| Butyricimonas                                 | 259                     | Parasutterella                              | 181                       |
| Phascolarctobacterium                         | 256                     | Phascolarctobacterium                       | 137                       |
| Parasutterella                                | 246                     | Butyricimonas                               | 104                       |
| Coprobacter                                   | 229                     | Desulfovibrio                               | 100                       |
| NK4A214 group                                 | 216                     | NK4A214 group                               | 92                        |

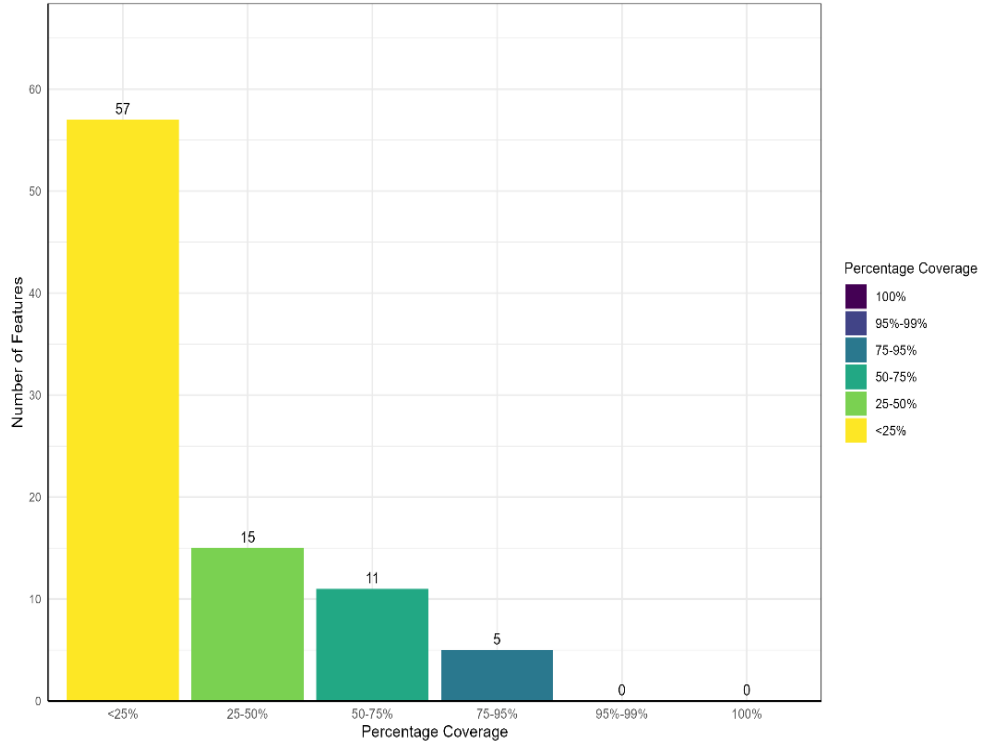
| <b>Metagenomic Species-Level</b> | <b>Max = 108</b> | <b>Metagenomic Species-Level</b> | <b>Max = 108</b> |
|----------------------------------|------------------|----------------------------------|------------------|
| Bacteroides faecis               | 96               | Gemmiger unknown_species         | 45               |
| Phocaeicola vulgatus             | 96               | Barnesiella intestinihominis     | 37               |
| Bacteroides caccae               | 92               | Phocaeicola vulgatus             | 30               |
| Alistipes communis               | 89               | Faecalibacterium unknown_species | 28               |
| Ruminococcus unknown_species     | 88               | Bacteroides caccae               | 22               |

Table 3: Top Five Most Frequently Occurring Features and Most Frequently Significant Features Across All Models for Each Taxonomic Level

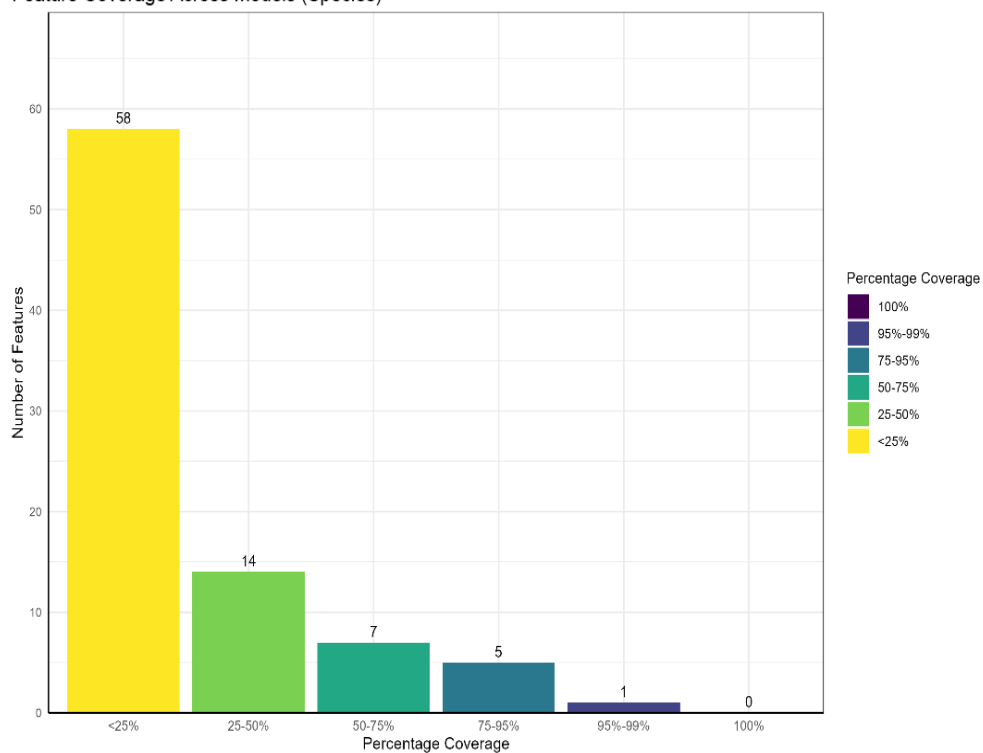
Feature Coverage Across Models (RSV)



Feature Coverage Across Models (Genus)



Feature Coverage Across Models (Species)



Feature Coverage Across Models (Metagenomic Species)

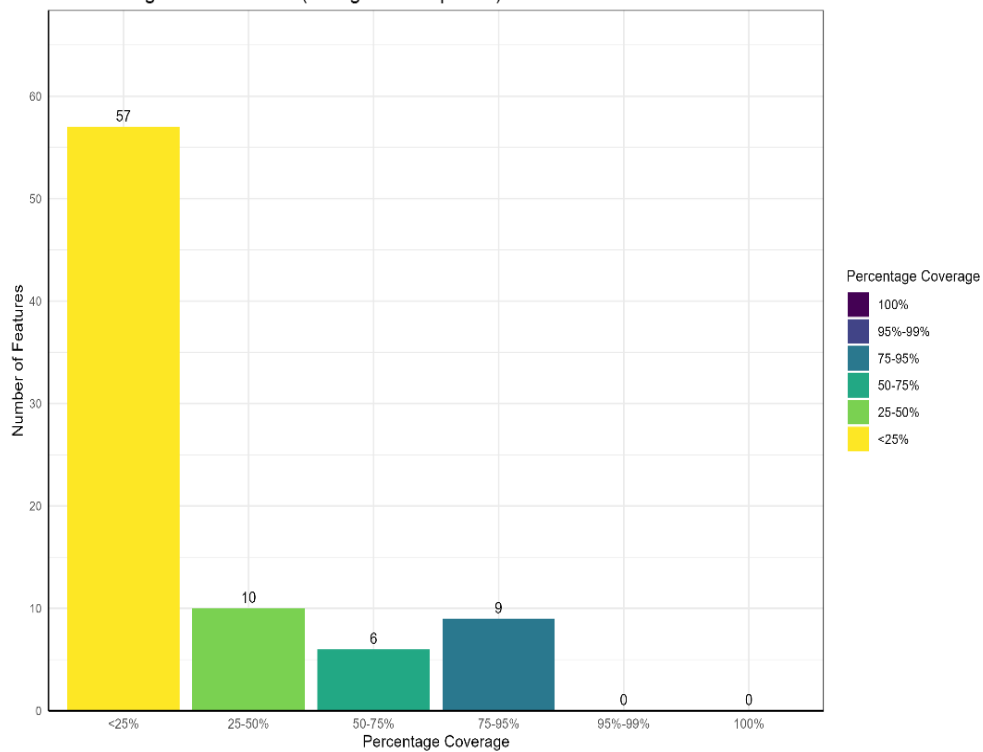
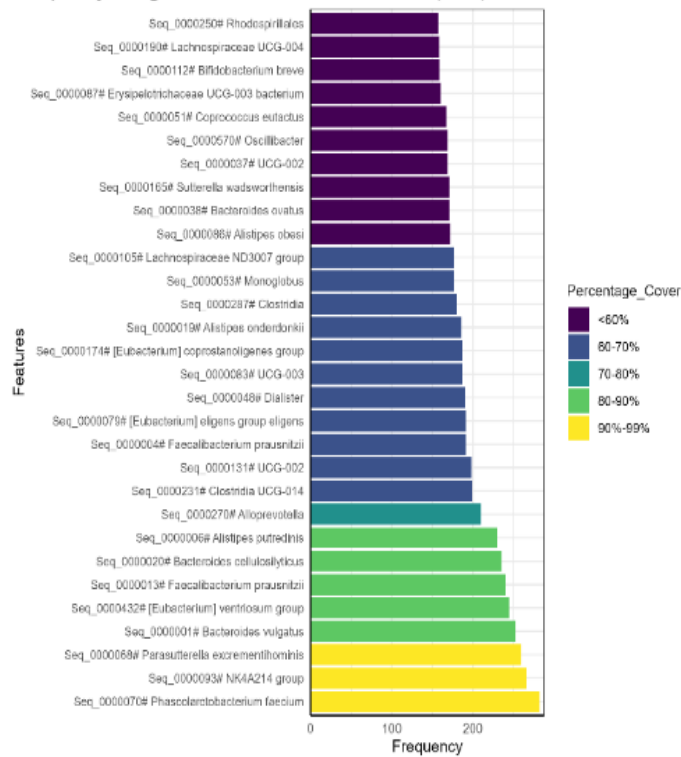


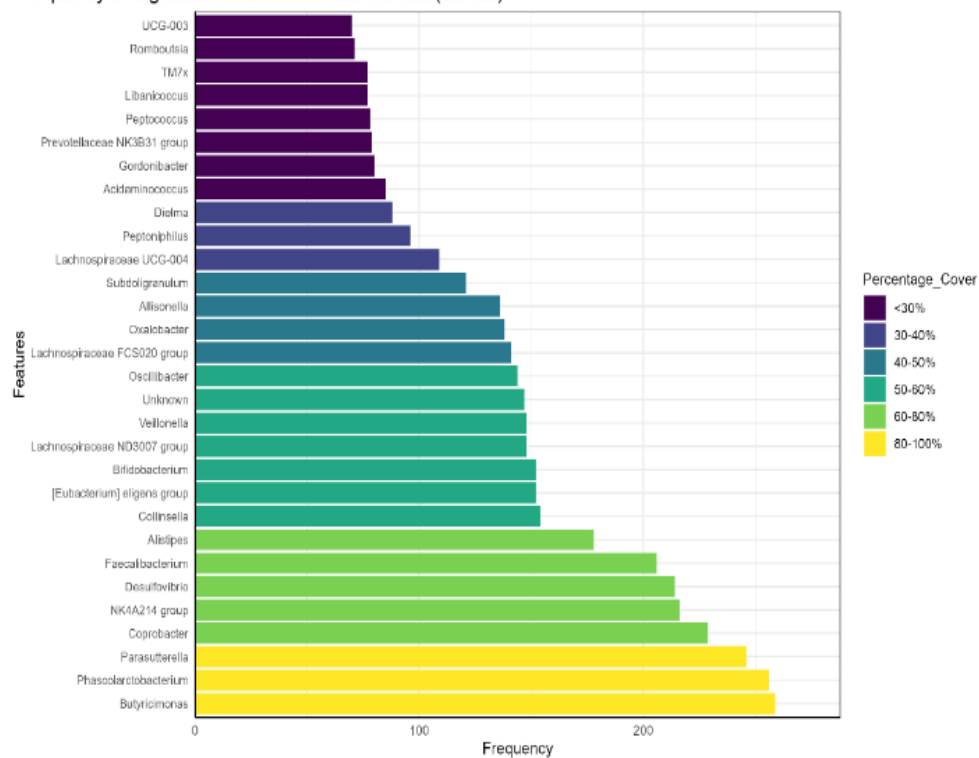
Figure 4a and 4b: Feature coverage of different taxonomic levels across models. The feature coverage is shown for RSVs, Species, Genus, and Metagenomic Species across various percentage coverage ranges. Each bar chart represents the number of features present within a certain percentage of the models trained. In all taxonomic levels, the majority of features have coverage below 25%, as indicated by the yellow bars. At the RSV level, 547 features have <25% coverage, while only 1 feature reaches 95–99% coverage. Similarly, for Species, Genus, and Metagenomic Species, the highest number of features fall in the <25% coverage group (57, 58, and 57 features, respectively). The number of features decreases significantly as the coverage percentage increases. The colour gradient represents the percentage coverage range.

Utilising the feature importance from the model collections, statistically significant ( $p\text{-value} \leq 0.05$ ) features were extracted and both their total frequency and significance frequency across the entire model collection was plotted to visualise their relevance (Figure 5 and 6). The five most dominant microbial species were all found to be statistically significant and similar observations were found for the other taxonomic levels (Figure 5). In total, at the RSV level, there were a total of 225 RSVs found to be significantly important for differentiation between sedentary controls and athletes (Supplementary table 1 of Appendix A). In contrast, there was a notable drop in the number of significant features at the species and genus levels, with 38 and 39 features, respectively (Supplementary table 2 and 3 of Appendix A). As such, the feature coverage for the RSV level was predominantly above 60% for all significant features. However, at both the species and genus levels, the majority of significant features exhibited coverage below 60%, with only 10 and 8 features surpassing this threshold, respectively. The metagenomic species data shared a similar structure to that of the 16S rDNA species data, with only 30 significant features being found and most of the significant feature's coverage being below 60% with only 12 features surpassing the threshold (Figure 5).

Frequency of Significant Features Across Models (RSV)



Frequency of Significant Features Across Models (Genus)



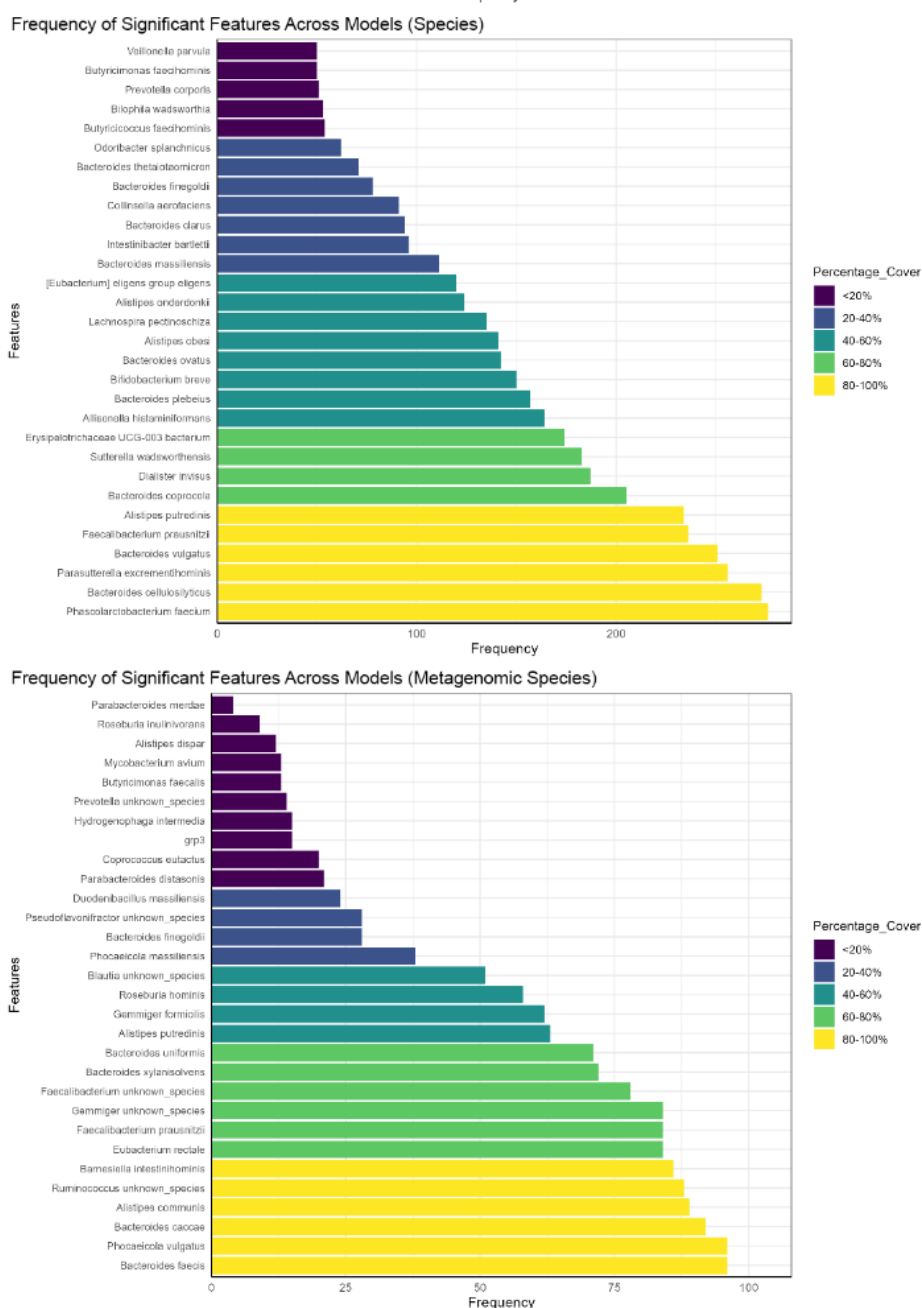
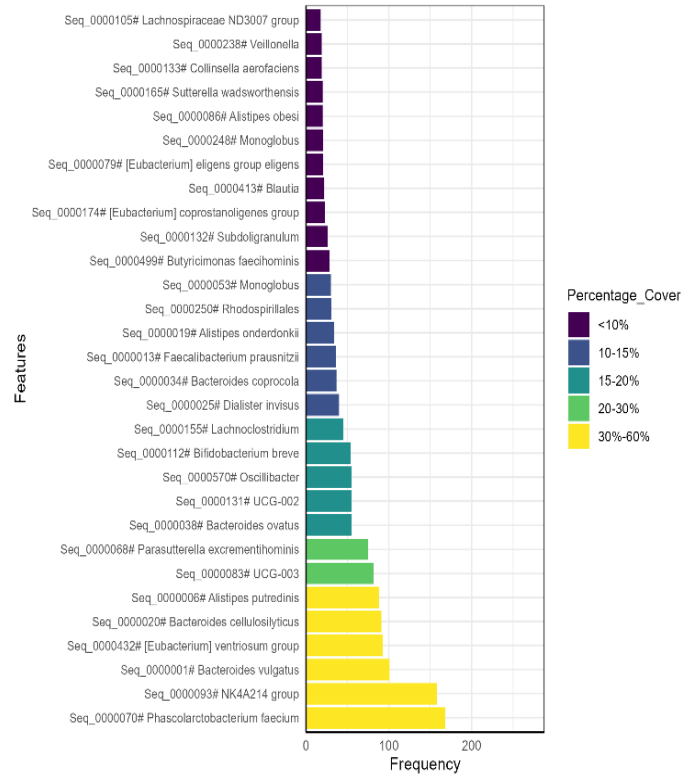
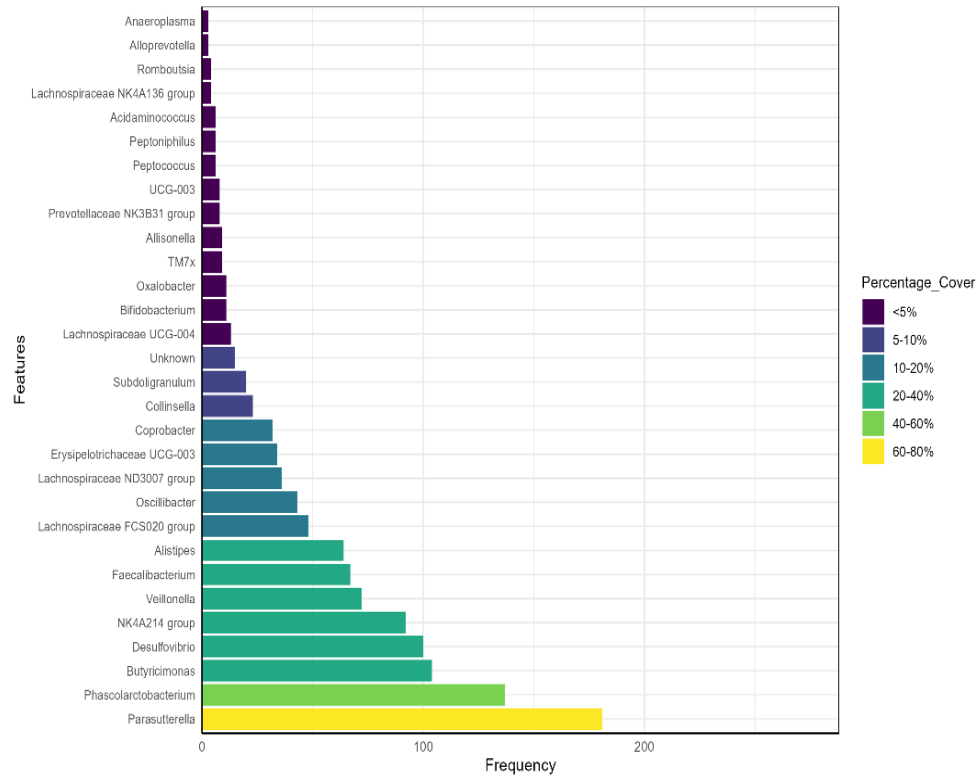


Figure 5a and 5b: Frequency Of The Top 30 Significant Features Across All Models For Each Taxonomic Level. Each bar chart illustrates the top 30 features and their respective frequencies across all models for RSV, Species, Genus, and Metagenomic Species. Across all taxonomic levels, except RSV, the majority of features exhibit a frequency of occurrence below 60%. In contrast, RSV predominantly displays features with a frequency above 60%. The bars are color-coded based on the percentage coverage of each feature, allowing for a visual comparison of feature prevalence within each taxonomic level across the models.

Frequency of Significance for Significant Features Across Models (RSV)



Frequency of Significance for Significant Features Across Models (Genus)



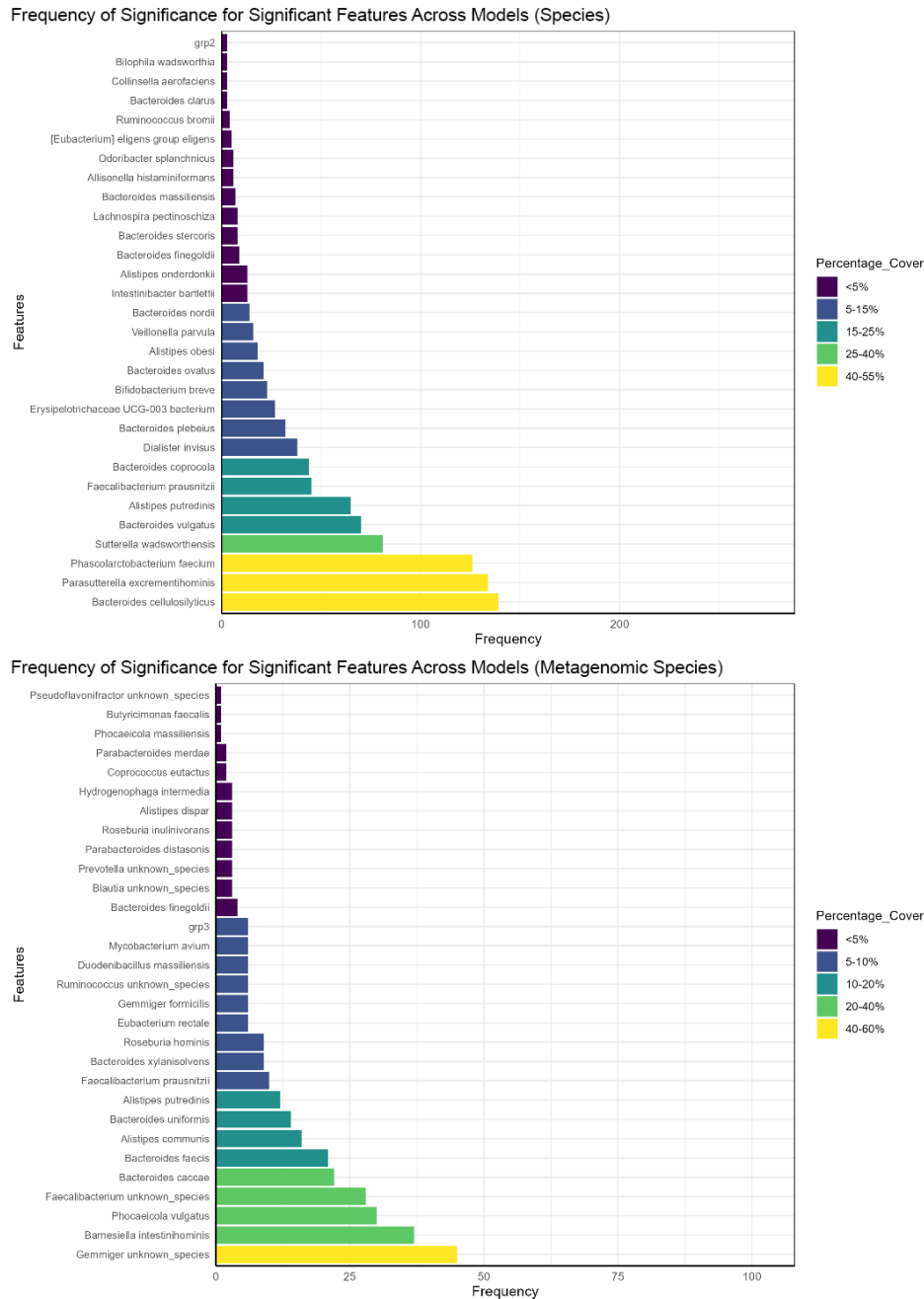


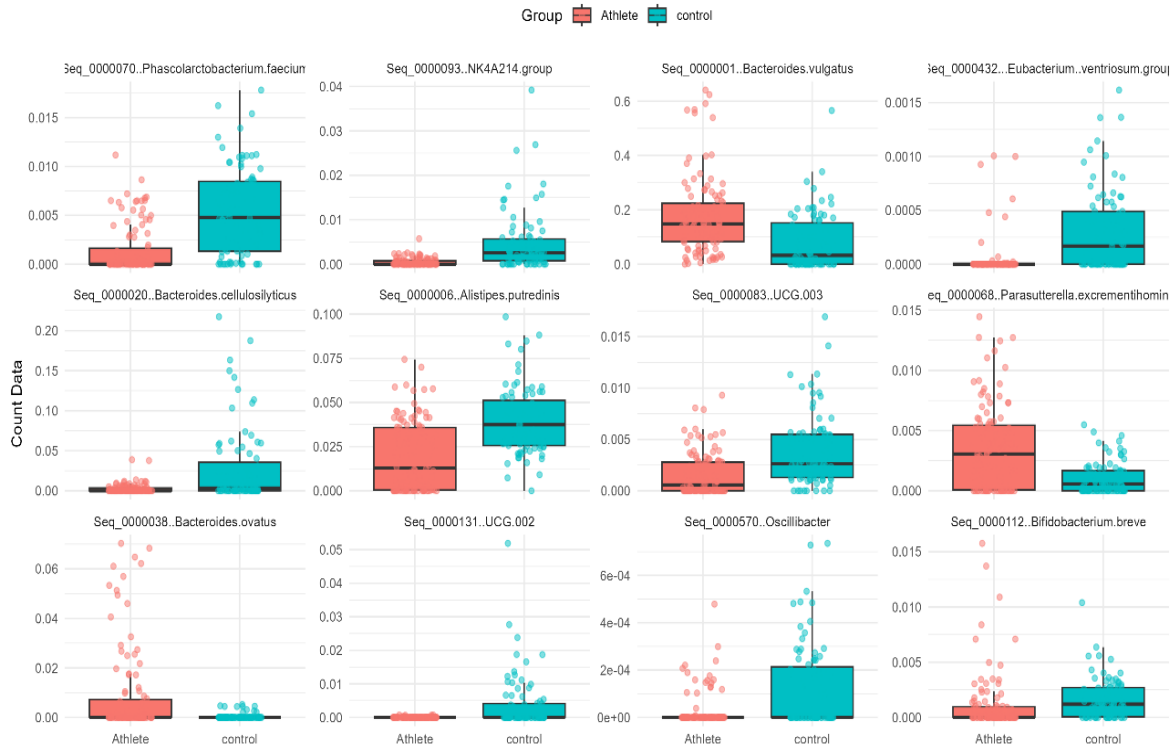
Figure 6a and 6b: Frequency Of Significance For The Top 30 Significant Features Across All Models For Each Taxonomic Level. Each bar chart illustrates the top 30 features and their respective frequency of significance across all models for RSV, Species, Genus, and Metagenomic Species. The bars are color-coded based on the percentage coverage of each feature, allowing for a visual comparison of feature prevalence within each taxonomic level across the models.

A similar approach was used to analyse how frequently each significant feature was found to be significant across the model collection (Figure 6). A substantial drop-off in the frequency of significance was observed for all features, and the ranking of dominant microbial species shifted from the earlier prevalence-based analysis. For example, *Phascolarctobacterium faecium*, which was previously the most dominant microbial species with a frequency of 276, now only has a significance frequency of 126, making it the third most dominant species in terms of significance (Table 3). This suggests that while *Phascolarctobacterium faecium* may be the most prevalent species, its role in distinguishing between sedentary controls and athletes is less significant than initially assumed.

The most significant species, based on frequency of significance, are *Bacteroides cellulosilyticus* (139), *Parasutterella excrementihominis* (134), *Phascolarctobacterium faecium* (126), *Sutterella wadsworthensis* (81), and *Bacteroides vulgatus* (70). A similar trend was observed in the metagenomic data, with the most significant species being *Gemmiger unknown\_species* (45), *Barnesiella intestinhominis* (37), *Phocaeicola vulgatus* (30), *Faecalibacterium unknown\_species* (28), *Bacteroides caccae* (22) (Table 3).

Lastly, the most significant features (based on frequency of significance) and their relative abundance data were used to highlight the difference in microbial composition between the sedentary control and athlete groups. Boxplots were generated to visually compare the distribution of these features between the two groups, revealing distinct shifts in the median values and interquartile ranges for most species (Figure 7). This visualisation helps clarify why the predictive model may have prioritised certain features as important, especially those that exhibit clear differences in abundance but are not immediately intuitive in their biological relevance.

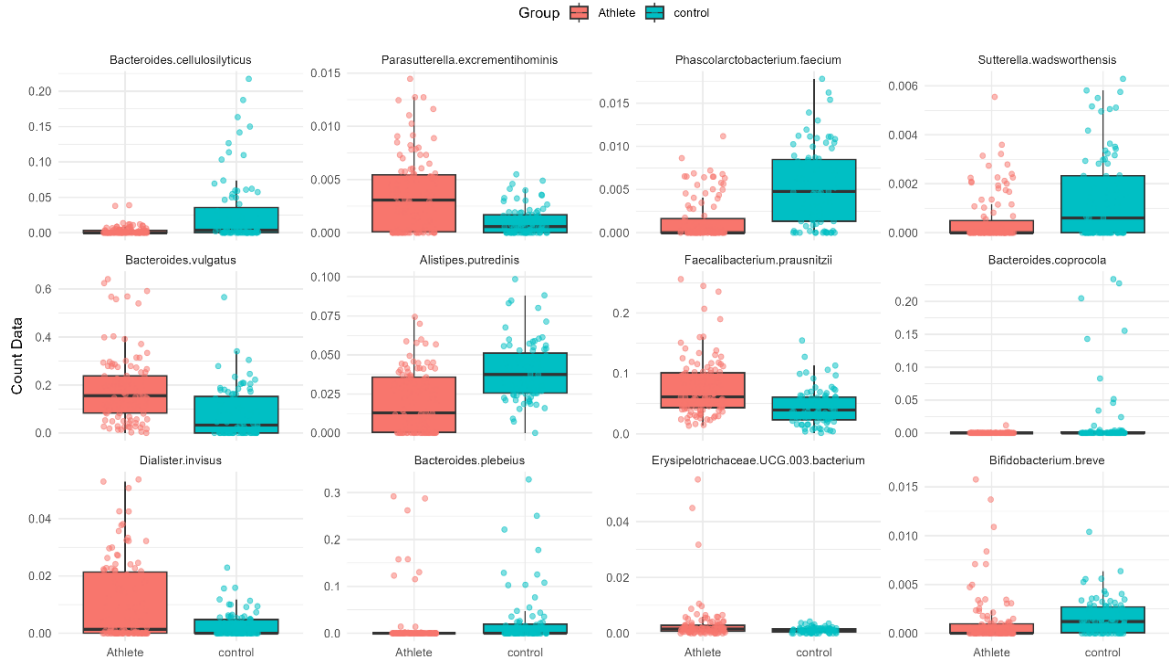
### Comparison of Control vs. Athlete Count Data for Most Significant RSV



### Comparison of Control vs. Athlete Count Data for Most Significant Genus



### Comparison of Control vs. Athlete Count Data for Most Significant Species



Comparison of Control vs. Athlete Count Data for Most Significant Metagenomic Species



Figure 7a and 7b: Comparison of microbial composition between sedentary control and athlete groups based on the most significant features. Boxplots illustrate the relative abundance distributions of the top significant microbial species between the two groups. Clear shifts in the median values and interquartile ranges are observed, emphasising the distinct differences in microbial composition. These differences help explain why the predictive models prioritised certain features as important.

To quantify these differences, a t-test was conducted to compare the mean abundance of each species between the control and athlete groups for both the 16S rDNA and metagenomic data. Out of the twelve species analysed from the 16S rDNA data, ten were found to have significantly different means ( $p < 0.05$ ), indicating strong evidence of microbial composition differences between the two groups. Notably, *Parasutterella excrementihominis* ( $p = 1.42\text{e-}10$ ), *Phascolarctobacterium faecium* ( $p = 4.02\text{e-}11$ ), and *Alistipes putredinis* ( $p = 3.43\text{e-}11$ ) exhibited the most significant differences. On the other hand, two species, *Bacteroides plebius* ( $p = 0.18$ ) and *Bifidobacterium breve* ( $p = 0.13$ ), were not significantly different between the groups (Table 4). Similar trends were observed at both the RSV and genus levels, where most features showed significant differences between sedentary controls and athletes. However, a few features were not significantly different (Table 4).

| <b>RSV-Level Significant Features</b>         | <b>P-Value</b>  | <b>Species-Level Significant Features</b>             | <b>P-Value</b>  |
|-----------------------------------------------|-----------------|-------------------------------------------------------|-----------------|
| Seq_0000070# Phascolarctobacterium faecium    | 4.02E-11        | Bacteroides cellulosilyticus                          | 7.71E-06        |
| Seq_0000093# NK4A214 group                    | 1.66E-07        | Parasutterella excrementihominis                      | 1.42E-10        |
| Seq_0000001# Bacteroides vulgatus             | 1.34E-07        | Phascolarctobacterium faecium                         | 4.02E-11        |
| Seq_0000432# [Eubacterium] ventriosum group   | 1.44E-07        | Sutterella wadsworthensis                             | 1.08E-04        |
| Seq_0000020# Bacteroides cellulosilyticus     | 7.71E-06        | Bacteroides vulgatus                                  | 2.66E-08        |
| Seq_0000006# Alistipes putredinis             | 3.43E-11        | Alistipes putredinis                                  | 3.43E-11        |
| Seq_0000083# UCG 003                          | 2.99E-07        | Faecalibacterium prausnitzii                          | 5.59E-08        |
| Seq_0000068# Parasutterella excrementihominis | 1.42E-10        | Bacteroides coprocola                                 | 5.80E-03        |
| Seq_0000038# Bacteroides ovatus               | 2.73E-06        | Dialister invisus                                     | 3.47E-07        |
| Seq_0000131# UCG 002                          | 5.46E-05        | Bacteroides plebeius                                  | <b>1.81E-01</b> |
| Seq_0000570# Oscillibacter                    | 1.57E-05        | Erysipelotrichaceae UCG 003 bacterium                 | 4.82E-03        |
| Seq_0000112# Bifidobacterium breve            | <b>1.27E-01</b> | Bifidobacterium breve                                 | <b>1.27E-01</b> |
| <b>Genus-Level Significant Features</b>       | <b>P-Value</b>  | <b>Metagenomic Species-Level Significant Features</b> | <b>P-Value</b>  |
| Parasutterella                                | 1.55E-10        | Gemmiger unknown_species                              | 2.85E-29        |
| Phascolarctobacterium                         | 4.06E-11        | Barnesiella intestinihominis                          | 3.64E-10        |
| Butyricimonas                                 | 1.02E-07        | Phocaeicola vulgatus                                  | 9.06E-11        |
| Desulfovibrio                                 | 3.29E-05        | Faecalibacterium unknown_species                      | 1.33E-21        |
| NK4A214 group                                 | 3.98E-06        | Bacteroides caccae                                    | 1.49E-08        |
| Veillonella                                   | 1.06E-04        | Bacteroides faecis                                    | 2.40E-07        |
| Faecalibacterium                              | 4.47E-08        | Alistipes communis                                    | 2.48E-07        |
| Alistipes                                     | 4.19E-09        | Bacteroides uniformis                                 | 1.49E-11        |
| Lachnospiraceae FCS020 group                  | 5.63E-05        | Alistipes putredinis                                  | 3.86E-08        |
| Oscillibacter                                 | 2.02E-04        | Faecalibacterium prausnitzii                          | 2.00E-18        |
| Lachnospiraceae ND3007 group                  | 9.98E-06        | Bacteroides xylanisolvens                             | 4.34E-06        |
| Erysipelotrichaceae UCG 003                   | 4.82E-03        | Roseburia hominis                                     | 4.69E-08        |

Table 4: Significance of Relative Abundance Between Control and Athlete Groups for the 12 Most Significant Features From Each Taxonomic Level

Whereas, for the metagenomic data, out of the twelve species analysed, all species were found to have significantly different means ( $p < 0.05$ ), indicating strong evidence of microbial composition differences between the two groups. Notably, *Gemmiger unknown\_species* ( $p = 2.85e-29$ ), *Faecalibacterium unknown\_species* ( $p = 1.33e-21$ ), and *Faecalibacterium prausnitzii* ( $p = 2.00e-18$ ) exhibited the most significant differences (Table 4).

## DISCUSSION

There is increasing recognition of the connection between exercise, athletic performance, and the GM. It is also understood that physiological and metabolic processes are not defined by a single organism but result from the activity of diverse bacterial populations (Topçuoğlu *et al.*, 2020). Consequently, traditional statistical methods that focus on analysing individual bacterial species in isolation fail to capture the full scope of biological interactions within the GM. In contrast, ML models account for the complex interactions between various microbes and the GM as a whole, helping to identify links between the microbiome and athletic performance (Jiang *et al.*, 2022).

To improve our understanding of how ML can provide insights into the microbiome, this study developed a specialised ML framework to address the challenges associated with microbiome research. The findings from the analysis of this framework indicates that models such as Random Forest, GLM, and SVM, particularly when combined with advanced feature selection methods like Boruta wrapper or embedded GLM and effective preprocessing techniques like CLR, provide the best predictive performance in distinguishing between sedentary control and athlete groups. The success of these parameters is likely due to their complementary strength, precision and appropriateness for the application of microbiome data.

CLR outperformed other preprocessing methods in terms of median AUC, like TSS and rCLR because it better handles the compositional nature of microbiome data, avoids correlation assumptions between features, utilises a log transformation as opposed to a simple proportional scaling, and uses imputation methods that provide reasonable estimates for values below the detection limit (Aitchison, 1982; Quinn *et al.*, 2019). Similarly, mRMR is not an effective feature selection method, producing a lower median AUC value, as it fails to capture the complexities of the data. In contrast, Boruta and GLM account for both feature relevance and model performance, which likely enhances predictive accuracy (Bolón-Canedo, Sánchez-Marroño and Alonso-Betanzos, 2013; Elwali and Moussavi, 2021).

The relationship between gut microbiota and athletic performance is an emerging area of interest and this study has identified several microbial species as having different relative abundances in athletes compared to controls, suggesting they play roles in modulating metabolism and enhancing performance (Hughes and Holscher, 2021).

While certain species are highly influential in distinguishing between athlete and control groups, no single feature serves as a universal marker of athletic performance across all models. This highlights the complexity and individual variability in biological responses, a finding supported by

previous studies that identified multiple taxa or species with varying abundances between athletes and controls. (Rinninella *et al.*, 2019; Humińska-Lisowska *et al.*, 2024).

Multiple butyrate-producing species from the *Faecalibacterium* genus were consistently identified as significant across all models based on feature importance. These species were also found in higher relative abundance in athletes compared to controls. The butyrate produced by *Faecalibacterium* likely supports enhanced athletic performance by promoting gut health and increasing the bioavailability of energy substrates (Clark and Mach, 2017). The consistent importance of the *Faecalibacterium* genus was observed in both 16S rDNA and metagenomic datasets, highlighting its key role in supporting athletic performance through improved gut health and energy metabolism. This point is further reinforced by findings from Fontana *et al.*, which identified a broad distribution of SCFA producing microbial species statistically associated with athlete samples (Fontana *et al.*, 2023).

Among these, species such as *Eubacterium rectale*, *Faecalibacterium prausnitzii*, and other unclassified *Faecalibacterium* species were particularly prevalent. Other significant but less prevalent species identified include *Blautia wexlerae* and *Ruminococcus bromii*. However, within the machine learning classification of my study, only *Faecalibacterium* species consistently emerged as significant in differentiating athletes from controls, while the other species were rarely found significant. These discrepancies may suggest limitations in the clustering techniques used by Fontana *et al.*, potentially leading to an overrepresentation of the importance of certain species (Fontana *et al.*, 2023). However, the consistent identification of *Faecalibacterium* species across multiple studies, despite differences in methodologies, further highlights the crucial role this genus plays in enhancing athletic performance (Jang *et al.*, 2019; Mailing *et al.*, 2019; Mohr *et al.*, 2020).

Similarly, another butyrate producing species, *Bacteroides vulgatus*, was found to be significant for differentiating between athletes and sedentary controls based on relative abundance and is more prevalent in athletes according to the 16S rDNA data. However, in the metagenomic data, it is less prominent in athletes, possibly due to competition with other butyrate and propionate-producing species that dominate the athletic microbiome (Hanson and Owen, 2004; Xu *et al.*, 2023). This discrepancy may also suggest that the differences in detection between the 16S rDNA and metagenomic approaches may reflect methodological biases. Similarly, the competition among microbial species for metabolic niches could explain why species like *Bacteroides cellulosilyticus* and *Phascolarctobacterium faecium*, which also produce propionate and acetate, are more prominent in controls rather than athletes.

Another notable observation from the study is the identification of several *Bacteroides* species that were significant in differentiating between athletes and controls across all the models. Interestingly, all of these *Bacteroides* species, excluding *B. vulgatus*, were found in higher relative abundance in controls compared to athletes. This trend suggests that while *Bacteroides* species play a role in general gut metabolism, they may not be as beneficial for athletic performance as other microbial species.

The higher abundance of *Bacteroides* in controls could be explained by their primary metabolic functions. *Bacteroides* species are known to break down carbohydrates, converting them into primary metabolites such as acetate and propionate (Johnson, Moore and Moore, 1986). While these metabolites are important for basic metabolic functions, their relative inefficiency compared to secondary metabolite producers, like *Faecalibacterium* and *Parasutterella*, may be less advantageous for the high energy demands of athletes. Athletes tend to favour microbial species that produce more efficient energy substrates, such as butyrate and succinate, which are more directly involved in enhancing endurance and muscle metabolism (Scheiman *et al.*, 2019; Wang *et al.*, 2019). Therefore, the higher prevalence of *Bacteroides* species in controls likely reflects a gut microbiome better suited for general health and carbohydrate metabolism rather than the optimised energy production needed for athletic performance. This is supported by previous studies that have found lower abundance of *Bacteroides* associated with athletes compared to healthy controls (Mailing *et al.*, 2019; Hughes and Holscher, 2021; O'Brien *et al.*, 2022).

The findings from this study appear contradictory to those of Scheiman *et al.*, which reported that athlete microbiomes were characterised by distinct microbial compositions with elevated abundances of *Bacteroides* (Scheiman *et al.*, 2019). However, Scheiman's analysis focused on the genus level and did not examine individual species within the *Bacteroides* genus. This detail is important to explain the difference in the results between the two studies. For example, while *B. vulgatus* shows a higher relative abundance in athletes, other *Bacteroides* species, such as *B. cellulosilyticus* and *Bacteroides coprocola*, are more abundant in controls. When *Bacteroides* is assessed as a single genus, the finer details of species-specific roles and interactions are often obscured. Therefore, the elevated abundance of *B. vulgatus* in athletes could mask the lower combined abundance of other *Bacteroides* species in controls.

However, arguably one of the most significant findings of Scheiman *et al.*, the identification of the *Veillonella* as a significant genus in the athletic microbiome, was supported by this study (Scheiman *et al.*, 2019). Yet, while this study supports *Veillonella* as a key genus for differentiating athletes from sedentary controls, it does not rank it as the most significant genus. This dissimilarity

likely stems from the differing methods employed by the studies. Scheiman et al., used differential abundance analysis, which, while useful, may not be as detailed as machine learning classification (Scheiman et al., 2019). Differential abundance analysis focuses on the presence and abundance of specific taxa but does not account for the complex interactions between microbial species. Whereas machine learning classification offers a more detailed approach by considering the overall contributions of species within the GM. This enables the identification of species that might be underrepresented or overlooked in differential abundance analyses, such as *Parasutterella* and *Phascolarctobacterium*, which were found as the most significant genera in this study. However, it is important to remember that *Veillonella*, a butyrate-producing species was found significant in both the study by Scheiman et al., and this study (Scheiman et al., 2019). This further emphasises the importance of butyrate producing species in enhancing the athletic performance of individuals.

Similarly, *Alistipes* species, while typically butyrate producers, are less abundant in athletes (Morgan et al., 2020). This may indicate that in athletes, more efficient butyrate-producing species have replaced *Alistipes*, like *F. prausnitzii*. The efficiency of butyrate production appears to be a determinant in distinguishing the gut microbiomes of athletes from those of sedentary controls, with more efficient producers likely contributing to enhanced metabolic health and athletic performance. To my knowledge, no studies have reported a decreased abundance of *Alistipes* in athletes compared to controls, nor have they identified *Alistipes* as significant for differentiating between athlete and control groups.

Additionally, several other species have been identified as being associated with athletic performance, including *Parasutterella excrementihominis*, *Sutterella wadsworthensis*, and an unknown species from the *Grimmiger* genus. These species were found to significantly differentiate between athlete and control groups based on feature importance, with both *P. excrementihominis* and *S. wadsworthensis* showing a higher relative abundance in athletes compared to controls. However, previous research has only linked *P. excrementihominis* and *S. wadsworthensis* to athletic performance in one prior study (Jang et al., 2019). No previous studies have found a connection between the *Grimmiger* genus and athletic performance, suggesting that this species may represent an unexplored or novel factor in understanding the gut microbiome's role in athleticism (Mailing et al., 2019; Mohr et al., 2020; Hughes and Holscher, 2021; O'Brien et al., 2022).

## Conclusion

This study has highlighted the potential of machine learning models in dissecting the complex relationships between the gut microbiome and athletic performance. ML models such as Random Forest, GLM, and SVM, particularly when combined with preprocessing techniques like CLR and feature selection methods like Boruta and embedded GLM, provided the highest predictive accuracy in distinguishing between athlete and control groups. The analysis also reinforces the need to explore multiple parameter configurations to capture the true effectiveness of a model, as performance can vary considerably depending on the selected parameters. Future studies should aim to expand parameter optimisation further to continue investigating trends found in this data, improving model generalisation and to ensure that biologically meaningful patterns are captured in machine learning applications for microbiome analysis.

The study also highlighted that the gut microbiome's role in athletic performance is not defined by any single species or genus but rather by a dynamic microbial ecosystem most prominently dominated by species from the *Faecalibacterium* genus and other butyrate-producing species. Furthermore, the study's results align with and expand upon previous research in this field by not only confirming the same trends but also identifying novel species that have not been reported in earlier studies. Future studies should focus on further identifying unknown species and exploring their metabolic contributions in controlled environments. The interaction between gut microbiota species and their collective impact on athletic performance remains an important area for further investigation, especially concerning the efficiency of SCFA production and the metabolic pathways involved.

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# APPENDICES

## Appendix A – Raw Data

| RSV-Level Features                                 | Frequency | Percentage Cover |
|----------------------------------------------------|-----------|------------------|
| Seq_0000070# Phascolarctobacterium faecium         | 282       | 90%-99%          |
| Seq_0000093# NK4A214 group                         | 266       | 90%-99%          |
| Seq_0000068# Parasutterella excrementihominis      | 260       | 90%-99%          |
| Seq_0000001# Bacteroides vulgatus                  | 253       | 80-90%           |
| Seq_0000432# [Eubacterium] ventriosum group        | 245       | 80-90%           |
| Seq_0000013# Faecalibacterium prausnitzii          | 240       | 80-90%           |
| Seq_0000020# Bacteroides cellulosilyticus          | 235       | 80-90%           |
| Seq_0000006# Alistipes putredinis                  | 230       | 80-90%           |
| Seq_0000270# Alloprevotella                        | 210       | 70-80%           |
| Seq_0000231# Clostridia UCG-014                    | 199       | 60-70%           |
| Seq_0000131# UCG-002                               | 198       | 60-70%           |
| Seq_0000004# Faecalibacterium prausnitzii          | 191       | 60-70%           |
| Seq_0000079# [Eubacterium] eligens group eligens   | 191       | 60-70%           |
| Seq_0000048# Dialister                             | 190       | 60-70%           |
| Seq_0000083# UCG-003                               | 187       | 60-70%           |
| Seq_0000174# [Eubacterium] coprostanoligenes group | 187       | 60-70%           |
| Seq_0000019# Alistipes onderdonkii                 | 186       | 60-70%           |
| Seq_0000287# Clostridia                            | 180       | 60-70%           |
| Seq_0000053# Monoglobus                            | 177       | 60-70%           |
| Seq_0000105# Lachnospiraceae ND3007 group          | 177       | 60-70%           |
| Seq_0000086# Alistipes obesi                       | 172       | <60%             |
| Seq_0000038# Bacteroides ovatus                    | 171       | <60%             |
| Seq_0000165# Sutterella wadsworthensis             | 171       | <60%             |
| Seq_0000037# UCG-002                               | 169       | <60%             |
| Seq_0000570# Oscillibacter                         | 169       | <60%             |
| Seq_0000051# Coprococcus eutactus                  | 167       | <60%             |
| Seq_0000087# Erysipelotrichaceae UCG-003 bacterium | 160       | <60%             |
| Seq_0000112# Bifidobacterium breve                 | 159       | <60%             |
| Seq_0000190# Lachnospiraceae UCG-004               | 158       | <60%             |
| Seq_0000250# Rhodospirillales                      | 157       | <60%             |
| Seq_0000155# Lachnoclostridium                     | 156       | <60%             |
| Seq_0000243# Lachnospiraceae                       | 155       | <60%             |
| Seq_0000184# Parabacteroides gordonii              | 152       | <60%             |
| Seq_0000034# Bacteroides coprocola                 | 151       | <60%             |
| Seq_0000499# Butyricimonas faecihominis            | 150       | <60%             |
| Seq_0000025# Dialister invisus                     | 149       | <60%             |
| Seq_0000022# Bacteroides plebeius                  | 148       | <60%             |
| Seq_0000555# Allisonella histaminiformans          | 148       | <60%             |

|                                                           |     |      |
|-----------------------------------------------------------|-----|------|
| Seq_0000132# Subdoligranulum                              | 146 | <60% |
| Seq_0000029# Subdoligranulum                              | 145 | <60% |
| Seq_0000434# Coprobacter                                  | 143 | <60% |
| Seq_0000629# Firmicutes                                   | 142 | <60% |
| Seq_0000103# [Eubacterium] siraeum group                  | 141 | <60% |
| Seq_0000009# Bacteroides thetaiotaomicron                 | 137 | <60% |
| Seq_0000032# Blautia                                      | 130 | <60% |
| Seq_0000280# [Eubacterium] ventriosum group               | 130 | <60% |
| Seq_0000745# Ruminococcaceae                              | 130 | <60% |
| Seq_0000088# Oscillibacter                                | 128 | <60% |
| Seq_0000479# Ruminococcaceae                              | 125 | <60% |
| Seq_0000248# Monoglobus                                   | 123 | <60% |
| Seq_0000260# Subdoligranulum                              | 123 | <60% |
| Seq_0000272# Gastranaerophilales                          | 123 | <60% |
| Seq_0000163# Bacteroides                                  | 120 | <60% |
| Seq_0000415# Ruminococcaceae                              | 120 | <60% |
| Seq_0000413# Blautia                                      | 119 | <60% |
| Seq_0000240# Clostridia UCG-014                           | 114 | <60% |
| Seq_0000343# Lachnospiraceae                              | 112 | <60% |
| Seq_0000117# Subdoligranulum                              | 111 | <60% |
| Seq_0000133# Collinsella aerofaciens                      | 110 | <60% |
| Seq_0000632# Megasphaera massiliensis                     | 110 | <60% |
| Seq_0000291# Faecalibacterium                             | 109 | <60% |
| Seq_0000312# [Eubacterium] eligens group                  | 109 | <60% |
| Seq_0000069# Agathobacter faecis                          | 107 | <60% |
| Seq_0000238# Veillonella                                  | 106 | <60% |
| Seq_0000979# Acidaminococcus                              | 106 | <60% |
| Seq_0000058# Alistipes finegoldii                         | 105 | <60% |
| Seq_0000986# Lachnospiraceae ND3007 group                 | 102 | <60% |
| Seq_0000074# Bacteroides vulgatus                         | 101 | <60% |
| Seq_0000439# Oscillibacter                                | 101 | <60% |
| Seq_0000046# Ruminococcus bicirculans                     | 100 | <60% |
| Seq_0000125# Lachnospira pectinoschiza                    | 100 | <60% |
| Seq_0000449# Butyricimonas                                | 100 | <60% |
| Seq_0000183# Enterobacteriaceae                           | 99  | <60% |
| Seq_0000040# Akkermansia muciniphila                      | 98  | <60% |
| Seq_0000084# [Eubacterium] ventriosum group<br>ventriosum | 98  | <60% |
| Seq_0000094# Bacteroides vulgatus                         | 98  | <60% |
| Seq_0000527# Lachnoclostridium                            | 96  | <60% |
| Seq_0000790# Clostridia UCG-014                           | 96  | <60% |
| Seq_0000054# Faecalibacterium                             | 95  | <60% |
| Seq_0000061# Lachnospiraceae                              | 94  | <60% |

|                                                        |    |      |
|--------------------------------------------------------|----|------|
| Seq_0000099# Clostridia UCG-014                        | 94 | <60% |
| Seq_0000390# Blautia                                   | 94 | <60% |
| Seq_0000978# Oscillospiraceae                          | 92 | <60% |
| Seq_0000206# Bacteroides nordii                        | 90 | <60% |
| Seq_0000027# Bacteroides massiliensis                  | 89 | <60% |
| Seq_0000444# Lachnospiraceae FCS020 group<br>bacterium | 89 | <60% |
| Seq_0000538# Turicibacter sanguinis                    | 88 | <60% |
| Seq_0000383# Veillonella parvula                       | 86 | <60% |
| Seq_0001168# Intestinimonas                            | 86 | <60% |
| Seq_0000264# Coprobacter fastidiosus                   | 82 | <60% |
| Seq_0000810# Monoglobus                                | 82 | <60% |
| Seq_0000779# Ruminococcaceae                           | 81 | <60% |
| Seq_0001072# Lachnospiraceae                           | 80 | <60% |
| Seq_0001643# Clostridia                                | 80 | <60% |
| Seq_0002448# [Eubacterium] ventriosum group            | 80 | <60% |
| Seq_0000017# Bacteroides                               | 76 | <60% |
| Seq_0000145# UCG-005                                   | 74 | <60% |
| Seq_0000024# Bacteroides ovatus                        | 73 | <60% |
| Seq_0000274# UCG-003                                   | 73 | <60% |
| Seq_0000152# Butyricicoccus faecihominis               | 72 | <60% |
| Seq_0000794# Parabacteroides                           | 72 | <60% |
| Seq_0000437# CAG-56                                    | 70 | <60% |
| Seq_0000446# Oxalobacter formigenes                    | 68 | <60% |
| Seq_0000589# [Clostridium] innocuum group              | 67 | <60% |
| Seq_0000230# Faecalibacterium                          | 66 | <60% |
| grp3                                                   | 65 | <60% |
| Seq_0001425# UCG-010                                   | 65 | <60% |
| Seq_0001618# Prevotellaceae                            | 65 | <60% |
| grp12                                                  | 64 | <60% |
| grp15                                                  | 64 | <60% |
| grp16                                                  | 64 | <60% |
| grp18                                                  | 64 | <60% |
| grp5                                                   | 64 | <60% |
| grp6                                                   | 64 | <60% |
| Seq_0000030# UCG-002                                   | 64 | <60% |
| Seq_0000242# CAG-352                                   | 64 | <60% |
| Seq_0000095# Bifidobacterium                           | 63 | <60% |
| Seq_0000958# UCG-005                                   | 62 | <60% |
| Seq_0000096# Bilophila wadsworthia                     | 61 | <60% |
| Seq_0000735# Lachnospiraceae UCG-004                   | 60 | <60% |
| grp20                                                  | 55 | <60% |
| Seq_0000386# Barnesiella                               | 55 | <60% |

|                                                   |    |      |
|---------------------------------------------------|----|------|
| Seq_0000438# Bacteroides                          | 55 | <60% |
| Seq_0000122# Lachnospiraceae                      | 54 | <60% |
| Seq_0000600# Streptococcus                        | 54 | <60% |
| Seq_0000041# Lachnospiraceae NK4A136 group        | 53 | <60% |
| Seq_0001307# Erysipelatoclostridium               | 53 | <60% |
| Seq_0000109# Romboutsia ilealis                   | 52 | <60% |
| Seq_0000463# Lachnospiraceae UCG-010              | 52 | <60% |
| Seq_0001014# Lachnospiraceae                      | 52 | <60% |
| Seq_0000002# Bacteroides uniformis                | 51 | <60% |
| Seq_0000286# RF39                                 | 51 | <60% |
| Seq_0000689# Christensenellaceae R-7 group        | 51 | <60% |
| Seq_0000057# Lachnoclostridium edouardi           | 50 | <60% |
| Seq_0000170# [Ruminococcus] gauvreauii group      | 50 | <60% |
| Seq_0000198# Prevotellaceae                       | 50 | <60% |
| Seq_0000258# UBA1819                              | 50 | <60% |
| Seq_0000750# Dorea                                | 50 | <60% |
| Seq_0000299# GCA-900066575                        | 48 | <60% |
| Seq_0000384# Ruminococcaceae                      | 48 | <60% |
| Seq_0000942# Lachnospiraceae UCG-004              | 48 | <60% |
| Seq_0000055# UCG-002                              | 47 | <60% |
| Seq_0000259# Coprococcus                          | 46 | <60% |
| Seq_0000729# Colidextribacter                     | 46 | <60% |
| Seq_0002361# Eisenbergiella                       | 46 | <60% |
| Seq_0000381# Faecalibacterium prausnitzii         | 45 | <60% |
| grp22                                             | 44 | <60% |
| Seq_0000075# [Ruminococcus] torques group torques | 44 | <60% |
| Seq_0000278# Lachnospiraceae UCG-010              | 44 | <60% |
| Seq_0002441# UCG-005                              | 43 | <60% |
| Seq_0000059# Bacteroides finegoldii               | 42 | <60% |
| Seq_0000219# RF39                                 | 41 | <60% |
| Seq_0000339# Lachnospiraceae                      | 40 | <60% |
| Seq_0000436# Lachnospiraceae UCG-010              | 40 | <60% |
| Seq_0000454# Peptococcaceae                       | 40 | <60% |
| Seq_0000502# Alistipes                            | 40 | <60% |
| Seq_0000294# UCG-005                              | 39 | <60% |
| Seq_0000472# Lachnospiraceae                      | 38 | <60% |
| Seq_0000015# Faecalibacterium prausnitzii         | 37 | <60% |
| Seq_0000543# Porphyromonas                        | 37 | <60% |
| Seq_0000642# Butyrivibrio                         | 37 | <60% |
| Seq_0000138# [Eubacterium] ruminantium group      | 36 | <60% |
| Seq_0000171# Bacteroides                          | 34 | <60% |
| Seq_0001028# Streptococcus                        | 34 | <60% |
| Seq_0000317# Ruminococcaceae                      | 33 | <60% |

|                                                    |    |      |
|----------------------------------------------------|----|------|
| Seq_0000076# RF39                                  | 32 | <60% |
| Seq_0000134# [Eubacterium] coprostanoligenes group | 32 | <60% |
| Seq_0000584# Coprobacter secundus                  | 32 | <60% |
| grp25                                              | 31 | <60% |
| Seq_0000026# Bacteroides caccae                    | 31 | <60% |
| Seq_0000113# Blautia obeum                         | 31 | <60% |
| Seq_0000120# Bacteroidales                         | 31 | <60% |
| Seq_0001015# Cloacibacillus evryensis              | 31 | <60% |
| Seq_0000489# Mailhella                             | 30 | <60% |
| Seq_0001655# Prevotellaceae                        | 30 | <60% |
| Seq_0000205# Alistipes                             | 29 | <60% |
| Seq_0000385# Sutterella                            | 29 | <60% |
| Seq_0000620# Lachnospira                           | 29 | <60% |
| Seq_0000726# Alistipes                             | 29 | <60% |
| Seq_0000010# Bacteroides stercoris                 | 28 | <60% |
| Seq_0000518# Prevotellaceae Ga6A1 group            | 28 | <60% |
| grp29                                              | 27 | <60% |
| grp30                                              | 27 | <60% |
| Seq_0000123# Lachnoclostridium                     | 27 | <60% |
| Seq_0000173# Lachnospiraceae UCG-004               | 26 | <60% |
| Seq_0000222# NK4A214 group                         | 26 | <60% |
| Seq_0000524# Alistipes                             | 26 | <60% |
| Seq_0000008# Subdoligranulum                       | 24 | <60% |
| Seq_0000031# Roseburia intestinalis                | 24 | <60% |
| Seq_0001458# Anaerofilum                           | 24 | <60% |
| Seq_0000067# [Eubacterium] siraeum group siraeum   | 23 | <60% |
| Seq_0000128# [Eubacterium] coprostanoligenes group | 23 | <60% |
| Seq_0001012# Enterobacteriaceae                    | 23 | <60% |
| Seq_0000256# Oscillospiraceae                      | 22 | <60% |
| Seq_0001709# Cloacibacillus                        | 22 | <60% |
| Seq_0000003# Bacteroides dorei                     | 21 | <60% |
| Seq_0000412# UCG-005                               | 21 | <60% |
| Seq_0000127# Ruminococcus                          | 20 | <60% |
| Seq_0000320# Intestinibacter bartlettii            | 19 | <60% |
| Seq_0000115# Coprococcus comes                     | 16 | <60% |
| Seq_0000116# Rikenellaceae RC9 gut group           | 16 | <60% |
| Seq_0000185# [Eubacterium] hallii group hallii     | 16 | <60% |
| Seq_0000665# Lachnospiraceae                       | 16 | <60% |
| Seq_0000863# Fusicatenibacter                      | 16 | <60% |
| Seq_0001802# UCG-005                               | 16 | <60% |
| Seq_0000092# Christensenellaceae R-7 group         | 15 | <60% |
| Seq_0000331# Lachnospiraceae NK4A136 group         | 15 | <60% |
| Seq_0001231# Cetobacterium somerae                 | 15 | <60% |

|                                               |    |      |
|-----------------------------------------------|----|------|
| Seq_0000730# Oscillospiraceae                 | 14 | <60% |
| Seq_0000969# Lachnospira                      | 13 | <60% |
| Seq_0000342# Lachnoclostridium                | 12 | <60% |
| Seq_0000855# Lachnospiraceae UCG-004          | 12 | <60% |
| Seq_0001070# [Eubacterium] xylanophilum group | 12 | <60% |
| Seq_0000023# Roseburia inulinivorans          | 10 | <60% |
| Seq_0000638# Incertae Sedis                   | 10 | <60% |
| Seq_0000893# Christensenellaceae R-7 group    | 8  | <60% |
| Seq_0000288# Parabacteroides gordonii         | 7  | <60% |
| Seq_0000448# Alistipes                        | 7  | <60% |
| Seq_0000097# [Eubacterium] xylanophilum group | 6  | <60% |
| Seq_0000049# Bacteroides ovatus               | 5  | <60% |
| Seq_0000254# Lachnospiraceae                  | 5  | <60% |
| Seq_0000962# Alistipes                        | 5  | <60% |
| Seq_0000976# Bifidobacterium animalis         | 5  | <60% |
| Seq_0000261# Lachnospiraceae                  | 4  | <60% |
| Seq_0000677# Coprobacter                      | 2  | <60% |

Supplementary Table 1: Frequency table of all significant RSV-Level Features

| Species-Level Features                                     | Frequency | Percentage Cover |
|------------------------------------------------------------|-----------|------------------|
| <i>Phascolarctobacterium faecium</i>                       | 276       | 80-100%          |
| <i>Bacteroides cellulosilyticus</i>                        | 273       | 80-100%          |
| <i>Parasutterella excrementihominis</i>                    | 256       | 80-100%          |
| <i>Bacteroides vulgatus</i>                                | 251       | 80-100%          |
| <i>Faecalibacterium prausnitzii</i>                        | 236       | 80-100%          |
| <i>Alistipes putredinis</i>                                | 234       | 80-100%          |
| <i>Bacteroides coprocola</i>                               | 205       | 60-80%           |
| <i>Dialister invisus</i>                                   | 187       | 60-80%           |
| <i>Sutterella wadsworthensis</i>                           | 183       | 60-80%           |
| <i>Erysipelotrichaceae</i> UCG-003 bacterium               | 174       | 60-80%           |
| <i>Allisonella histaminiformans</i>                        | 164       | 40-60%           |
| <i>Bacteroides plebeius</i>                                | 157       | 40-60%           |
| <i>Bifidobacterium breve</i>                               | 150       | 40-60%           |
| <i>Bacteroides ovatus</i>                                  | 142       | 40-60%           |
| <i>Alistipes obesi</i>                                     | 141       | 40-60%           |
| <i>Lachnospira pectinoschiza</i>                           | 135       | 40-60%           |
| <i>Alistipes onderdonkii</i>                               | 124       | 40-60%           |
| [ <i>Eubacterium</i> ] <i>eligens</i> group <i>eligens</i> | 120       | 40-60%           |
| <i>Bacteroides massiliensis</i>                            | 111       | 20-40%           |
| <i>Intestinibacter bartlettii</i>                          | 96        | 20-40%           |
| <i>Bacteroides clarus</i>                                  | 94        | 20-40%           |
| <i>Collinsella aerofaciens</i>                             | 91        | 20-40%           |
| <i>Bacteroides finegoldii</i>                              | 78        | 20-40%           |
| <i>Bacteroides thetaiotaomicron</i>                        | 71        | 20-40%           |
| <i>Odoribacter splanchnicus</i>                            | 62        | 20-40%           |
| <i>Butyricicoccus faecihominis</i>                         | 54        | <20%             |
| <i>Bilophila wadsworthia</i>                               | 53        | <20%             |
| <i>Prevotella corporis</i>                                 | 51        | <20%             |
| <i>Butyricimonas faecihominis</i>                          | 50        | <20%             |
| <i>Veillonella parvula</i>                                 | 50        | <20%             |
| <i>Lachnospiraceae</i> NK4A136 group bacterium             | 48        | <20%             |
| <i>Bacteroides nordii</i>                                  | 47        | <20%             |
| <i>Alistipes shahii</i>                                    | 42        | <20%             |
| <i>Ruminococcus bicirculans</i>                            | 42        | <20%             |
| <i>Ruminococcus bromii</i>                                 | 33        | <20%             |
| grp2                                                       | 23        | <20%             |
| <i>Methylobacterium-Methylobacterium jeotgali</i>          | 22        | <20%             |
| <i>Bacteroides stercoris</i>                               | 15        | <20%             |

Supplementary Table 2: Frequency table of all significant Species-Level Features

| <b>Genus-Level Features</b>     | <b>Frequency</b> | <b>Percentage Cover</b> |
|---------------------------------|------------------|-------------------------|
| Butyricimonas                   | 259              | 80-100%                 |
| Phascolarctobacterium           | 256              | 80-100%                 |
| Parasutterella                  | 246              | 80-100%                 |
| Coprobacter                     | 229              | 60-80%                  |
| NK4A214 group                   | 216              | 60-80%                  |
| Desulfovibrio                   | 214              | 60-80%                  |
| Faecalibacterium                | 206              | 60-80%                  |
| Alistipes                       | 178              | 60-80%                  |
| Collinsella                     | 154              | 50-60%                  |
| [Eubacterium] eligens group     | 152              | 50-60%                  |
| Bifidobacterium                 | 152              | 50-60%                  |
| Lachnospiraceae ND3007 group    | 148              | 50-60%                  |
| Veillonella                     | 148              | 50-60%                  |
| Unknown                         | 147              | 50-60%                  |
| Oscillibacter                   | 144              | 50-60%                  |
| Lachnospiraceae FCS020 group    | 141              | 40-50%                  |
| Oxalobacter                     | 138              | 40-50%                  |
| Allisonella                     | 136              | 40-50%                  |
| Subdoligranulum                 | 121              | 40-50%                  |
| Lachnospiraceae UCG-004         | 109              | 30-40%                  |
| Peptoniphilus                   | 96               | 30-40%                  |
| Dielma                          | 88               | 30-40%                  |
| Acidaminococcus                 | 85               | <30%                    |
| Gordonibacter                   | 80               | <30%                    |
| Prevotellaceae NK3B31 group     | 79               | <30%                    |
| Peptococcus                     | 78               | <30%                    |
| Libanicoccus                    | 77               | <30%                    |
| TM7x                            | 77               | <30%                    |
| Romboutsia                      | 71               | <30%                    |
| UCG-003                         | 70               | <30%                    |
| Lachnospiraceae NK4A136 group   | 68               | <30%                    |
| Erysipelotrichaceae UCG-003     | 58               | <30%                    |
| Anaeroplasma                    | 54               | <30%                    |
| Alloprevotella                  | 21               | <30%                    |
| Mailhella                       | 19               | <30%                    |
| [Eubacterium] ruminantium group | 14               | <30%                    |
| Pseudoflavonifractor            | 13               | <30%                    |
| Blautia                         | 4                | <30%                    |
| Erysipelatoclostridium          | 2                | <30%                    |

Supplementary Table 3: Frequency table of all significant Genus-Level Features